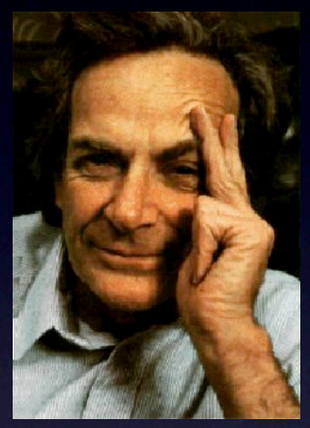
Symmetry and Quantum Algorithms

Dave Bacon
University of Washington

Department of
~~Computer Science & Engineering~~
Misfit Faculty



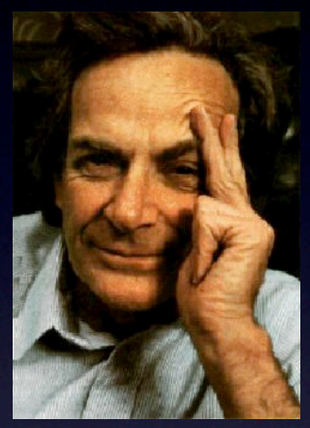
Quantum Propaganda



I think I can safely say that nobody understands quantum mechanics.

Richard Feynman
Nobel Prize 1965

Quantum Propaganda



I think I can safely say that nobody understands quantum mechanics.

Richard Feynman
Nobel Prize 1965



Anyone who is not shocked by quantum theory has not understood it.

Niels Bohr
Nobel Prize 1922

Hogwash



- They understand it
- They use it
- It just defies their sensibilities

These Are a Few of My Favorite Norms

p-norm: $||x||_p = \left[\sum_i |x_i|^p\right]^{1/p} \qquad x_i \in \mathbb{R}$

"0-norm" (not really a norm)

$$||x||_0 = \lim_{p \to 0} ||x||_p \qquad 0^0 \equiv 0$$

$$= \text{number of nonzero } x_i$$



1-norm (taxicab distance)

$$||x||_1 = \sum_i |x_i|$$

2-norm (euclidean distance)

$$||x||_2 = \sqrt{\sum_i |x_i|^2}$$

Deterministic

"Description" of n bits:

$$x = (x_0, x_1, \ldots, x_{2^n-1})$$

Enforce $||x||_0 = 1$

"Description" = vector with one nonzero entry

"Evolution" = matrix with one nonzero entry in every row and column

$$\begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 3 & 0 \\ 0 & 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 4 \end{bmatrix} \quad \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix}$$

"Observation" = bits in configuration with non-zero description entry

Probabilistic

"Description" of n bits:

$$x = (x_0, x_1, \ldots, x_{2^n-1})$$

Enforce $||x||_1 = 1$

"Description" = vector whose absolute values sum to unity, choice of sign forces a convention

"Evolution" = stochastic matrix (modulo the sign)

$$\begin{bmatrix} 0.5 & 0 & 0.6 & 0 \\ 0.25 & 1 & 0.4 & 0.7 \\ 0.1 & 0 & 0 & 0.1 \\ 0.15 & 0 & 0 & 0.2 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0.5 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.25 \\ 0.625 \\ 0.05 \\ 0.075 \end{bmatrix} \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix}$$

"Observation" = bits in configuration with probability given by the absolute value of entry in the description

Quantum

"Description" of n bits:

$$x = (x_0, x_1, \ldots, x_{2^n-1})$$

Enforce $||x||_2 = 1$

"Description" = vector whose absolute values squared sum to unity

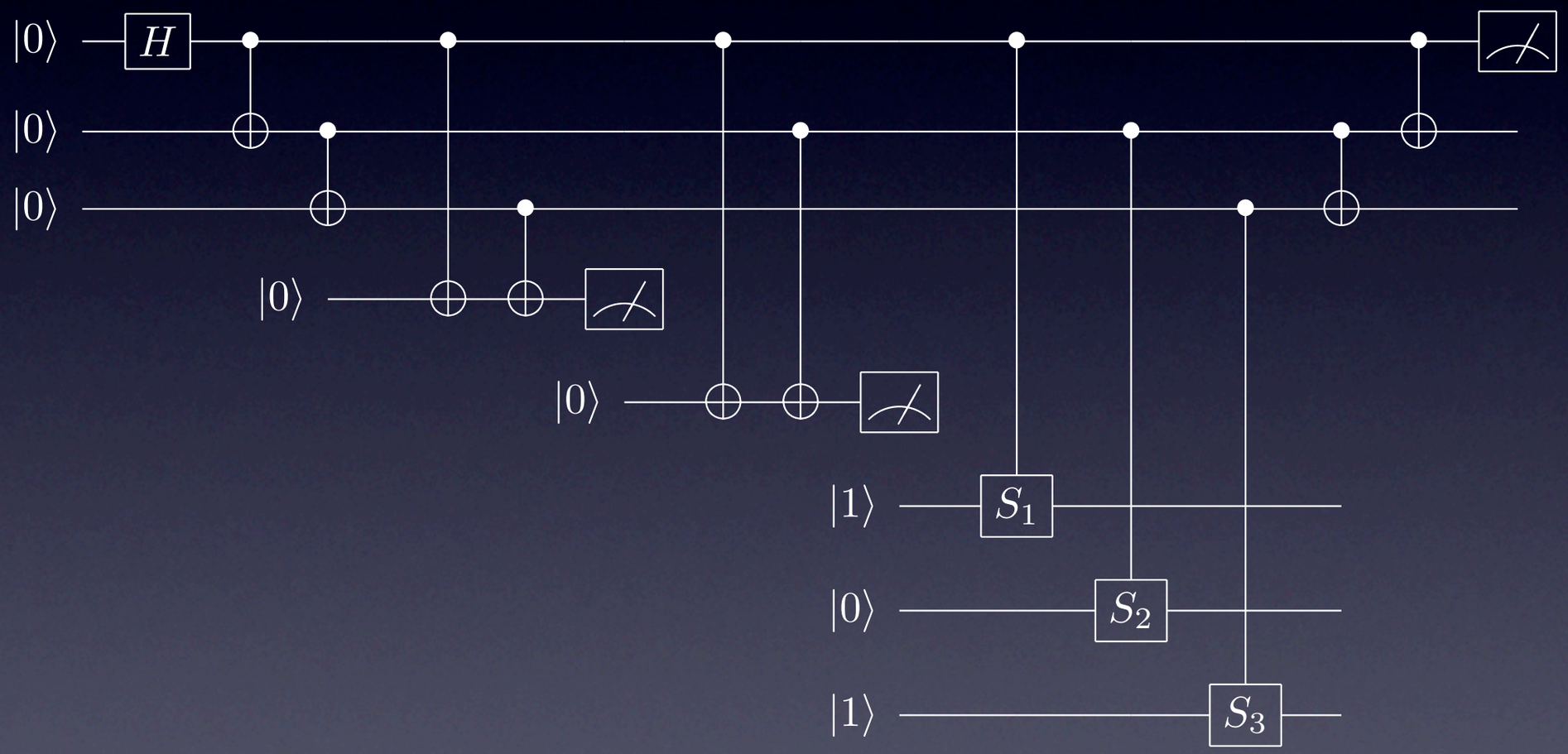
"Evolution" = orthogonal matrix

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix} \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix}$$

"Observation" = bits in configuration with probability given by the absolute value squared of entry in the description

Quantum Algorithms

Circuit model for quantum algorithms



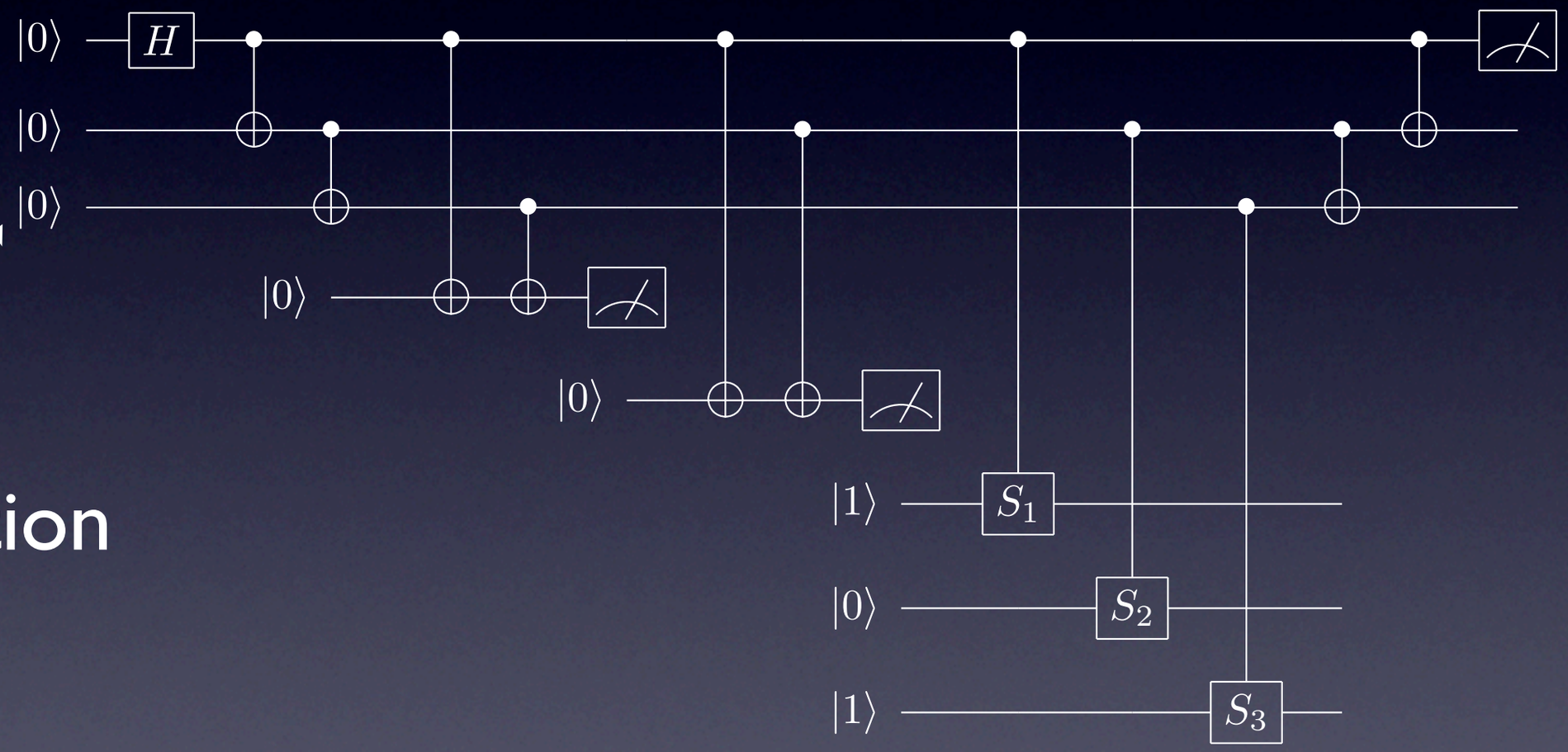
CS time ⟶

Quantum Algorithms

Circuit model for quantum algorithms

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

state preparation



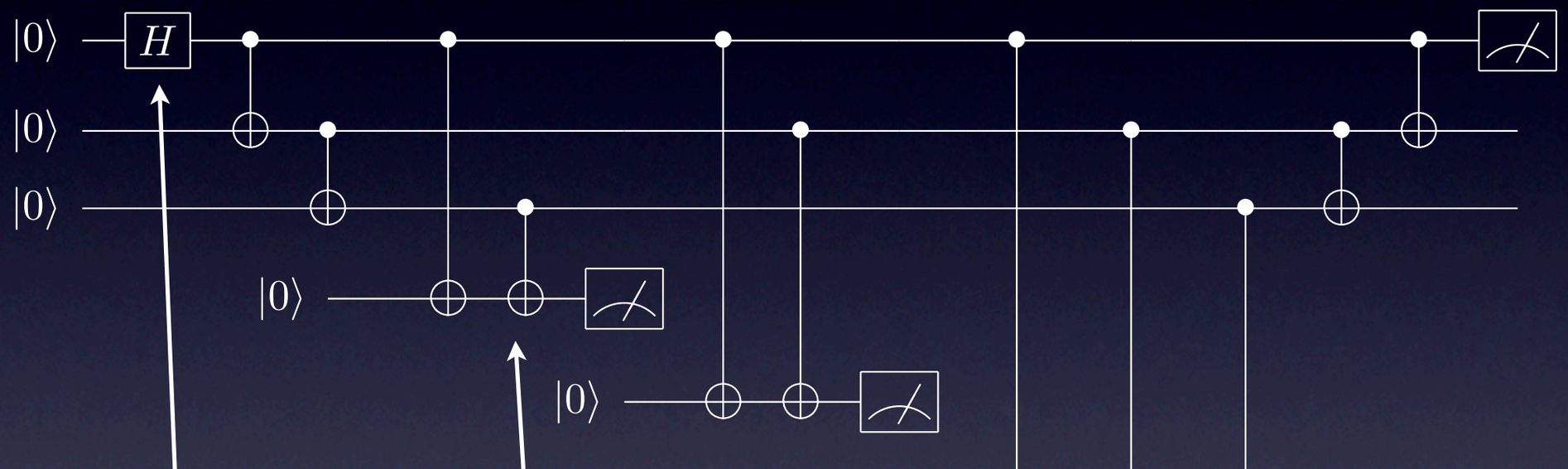
CS time ⟶

Quantum Algorithms

Circuit model for quantum algorithms

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state preparation



quantum gate

$$H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

quantum gate

$$C_X = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

CS time $\longrightarrow$

Quantum Algorithms

Circuit model for quantum algorithms

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

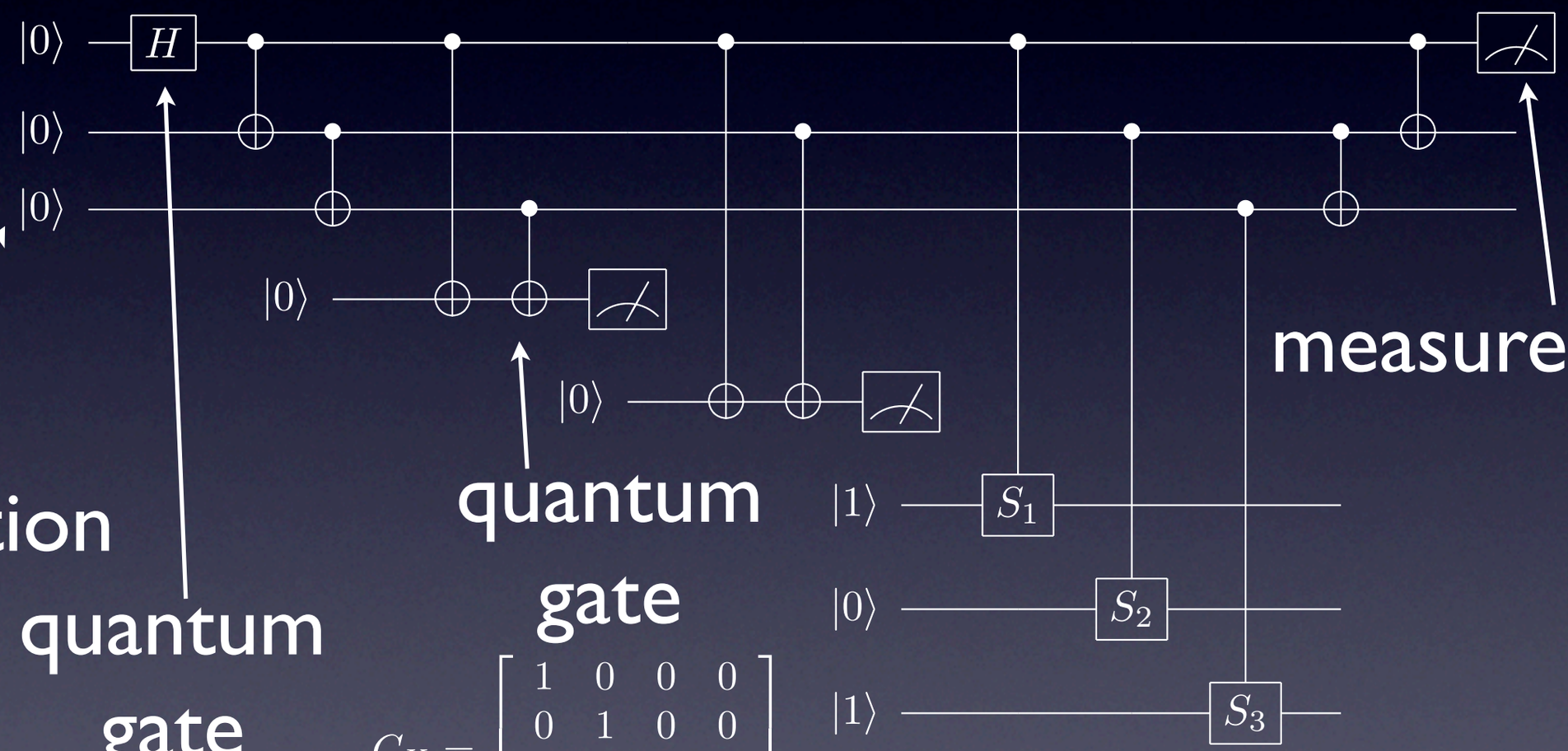
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quantum gate

$$C_X = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$



quantum gate

measurement

CS time ⟶

Physicist's Notation

$|\psi\rangle$ = column vector (ket)

$\langle\psi|$ = row vector (bra)

"computational basis":

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$\langle\psi|\phi\rangle$ = inner product (a number)

$|\phi\rangle\langle\psi|$ = outer product (a matrix)

Tensor product: $\begin{bmatrix} a \\ b \end{bmatrix} \otimes \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} ac \\ ad \\ bc \\ bd \end{bmatrix}$

Physicist's Notation II



Johnny von Neumann says

When you need to combine 1 and 2 norms, use the density matrix:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

$|\psi_i\rangle$ with probability p_i

Quantum Algorithms

Factor

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

Quantum Algorithms

Factor

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

=

398075086424064937397125500550386491199064362342526708406385189575946388957261768583317

x

472772146107435302536223071973048224632914695302097116459852171130520711256363590397527

Quantum Algorithms

Factor

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

=

398075086424064937397125500550386491199064362342526708406385189575946388957261768583317

x

472772146107435302536223071973048224632914695302097116459852171130520711256363590397527

	Running time
Best classical algorithm	$\exp(N^{1/3})$
Shor's 1994 quantum algorithm	N^3

Breaks public key cryptosystems (RSA, Elliptic curves)

Quantum Algorithms

Grover Type

Unstructured Search
[Grover 96]

Quantum amplitude amplification
[Brassard, Hoyer, Mosca, Tapp 00]

Adiabatic quantum algorithms
[Farhi, Goldstone Gutman, Sipser 00]

NAND tree algorithm
[Farhi, Goldstone Gutman 07]

Quantum simulated annealing
[Somma, Boixo, Barnum 07]

Misc Type

BQP completeness of Jones, Tutte, and other polynomials, Exponential speedups in quantum random walks, Simulation of Schrodinger's equation, Physical chemistry, etc.

Shor Type

Factoring, Discrete Logarithm
[Shor 94]

Pell's Equation
[Hallgren 02]

Some Hidden Shift Problems
[van Dam, Hallgren, Ip 02]
[van Dam, Childs 05]

Gauss Sums
[van Dam, Seroussi 02]

Hidden Subgroup Problems
Abelian groups, almost abelian groups, near Hamiltonian groups, normal subgroups, small commutator subgroup, solvable groups of bounded exponent and of bounded derived series extraspecial groups, groups of nilpotency class at most 2, more

On Quantum Algorithms

On Quantum Algorithms



My Nightmare →

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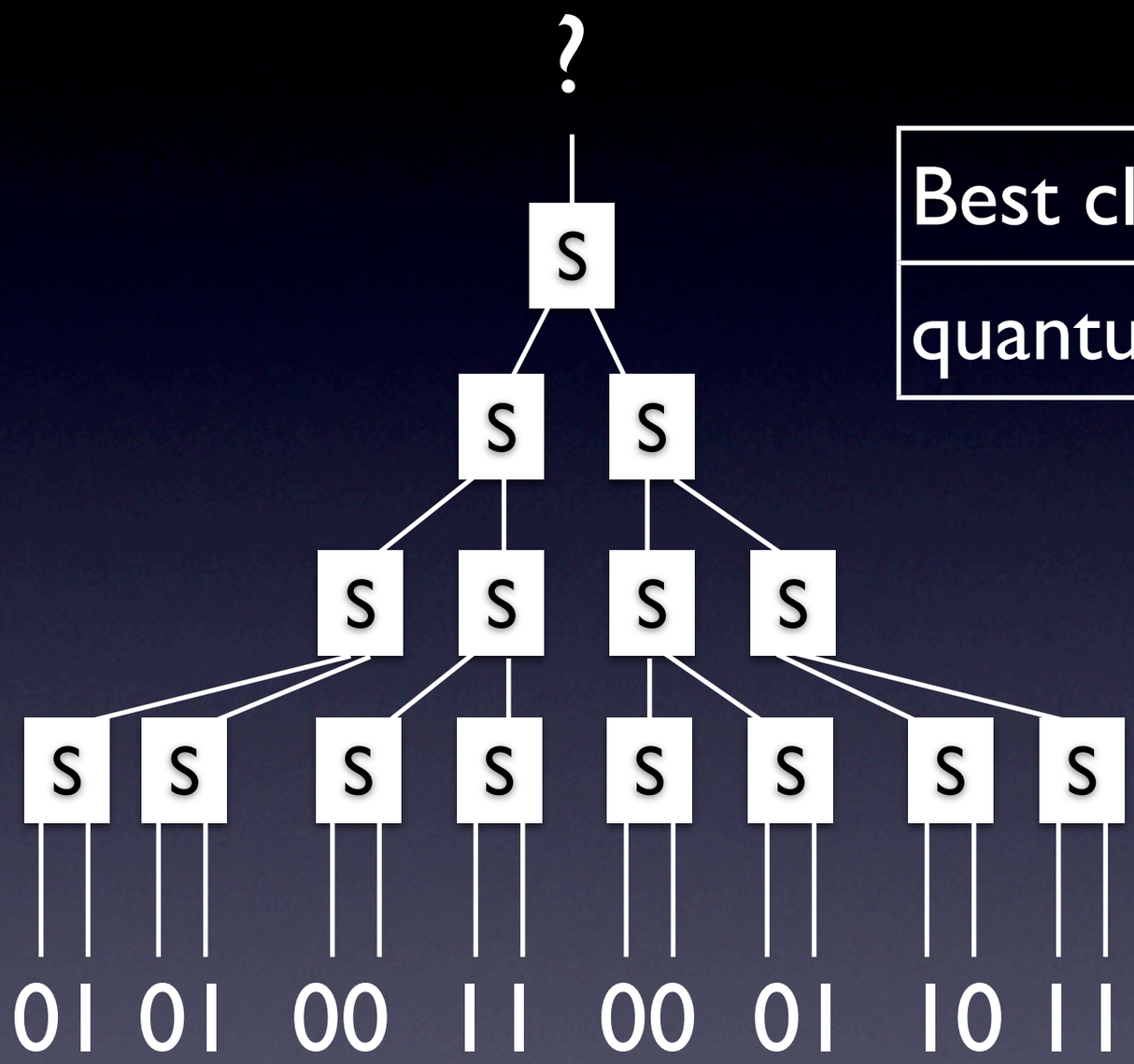
HELP WANTED Quantum programmer. Only skill necessary is ability to type in large RSA keys. Pay inversely commensurate with years since Ph.D. completed.

HELP WANTED Real estate agent. Best market since bubble popped in 08. Join a dedicated

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Quantum Algorithms



	Time
Best classical algorithm	N^a
quantum algorithm	$N^{1/2}$

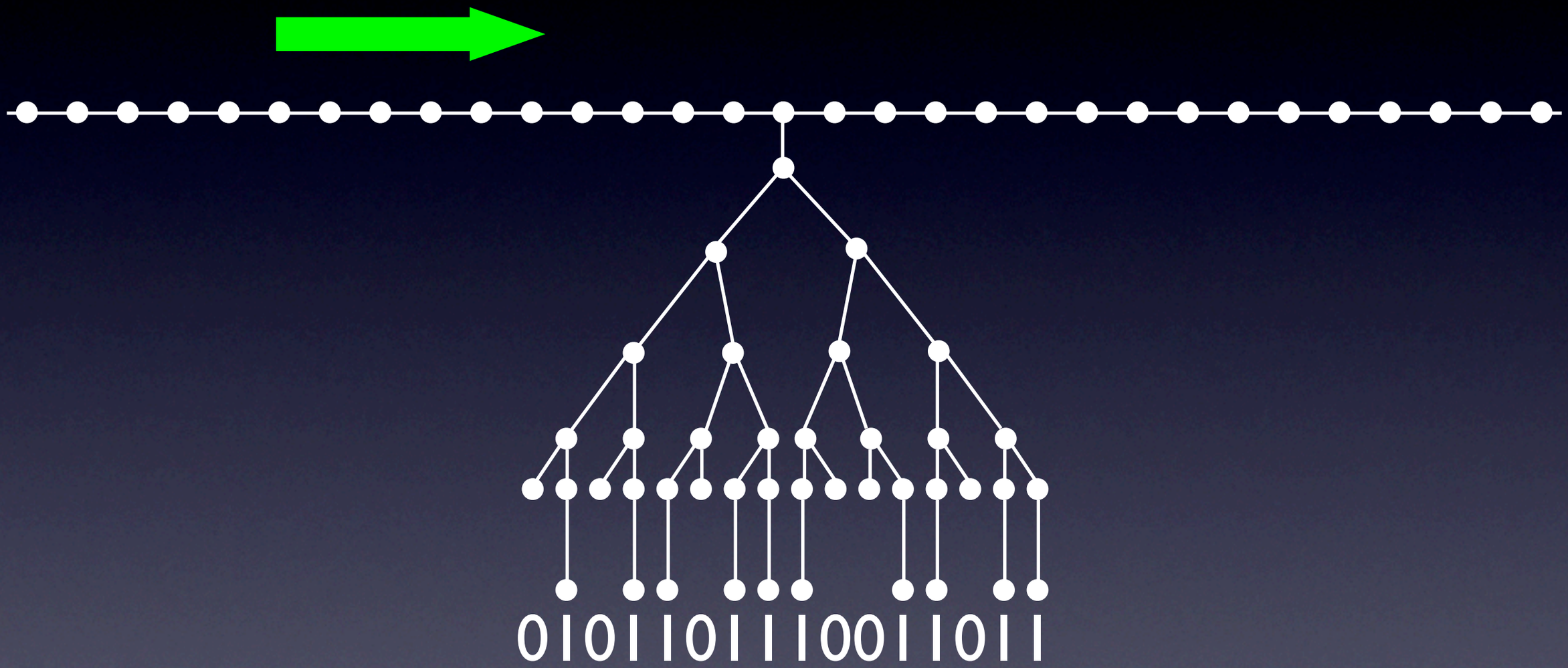
$$a = \log_2(1 + \sqrt{33}) - 2$$
$$a \approx 0.753\ldots$$

Find root by querying fewest leaves

Farhi, Goldstone, and Gutman, "A Quantum Algorithm for the Hamiltonian NAND Tree" 2007

Quantum Algorithms

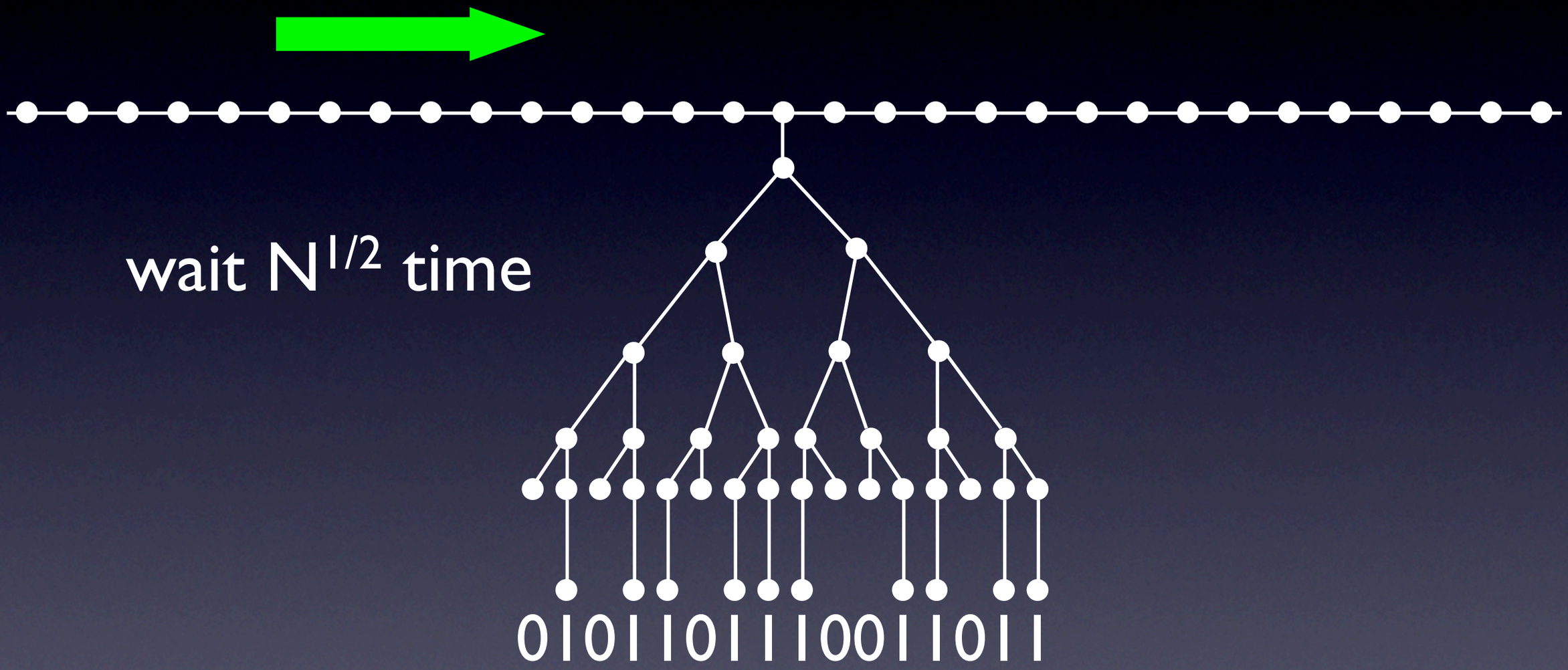
right going initial wave packet



Farhi, Goldstone, and Gutman, "A Quantum Algorithm for the Hamiltonian NAND Tree" 2007

Quantum Algorithms

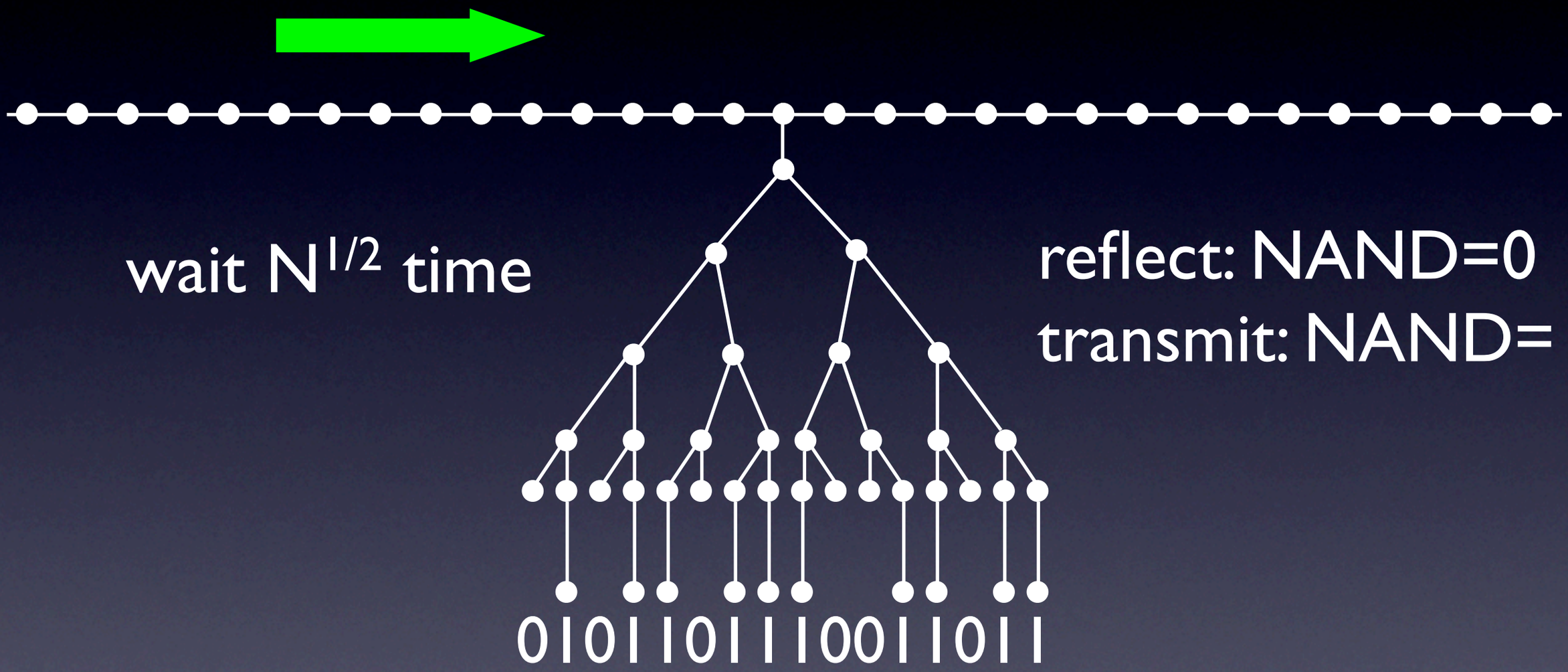
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Farhi, Goldstone, and Gutman, "A Quantum Algorithm for the Hamiltonian NAND Tree" 2007

Quantum Algorithms

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Farhi, Goldstone, and Gutman, "A Quantum Algorithm for the Hamiltonian NAND Tree" 2007

Questions

- Where does the power of quantum algorithms come from?

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- Where does the power of classical algorithms come from?

Questions

- Where does the power of ~~quantum algorithms come~~ from?
- Where does the power of ~~classical algorithms come~~ from?
- Why are quantum algorithms sometimes exponentially more powerful than classical computers?

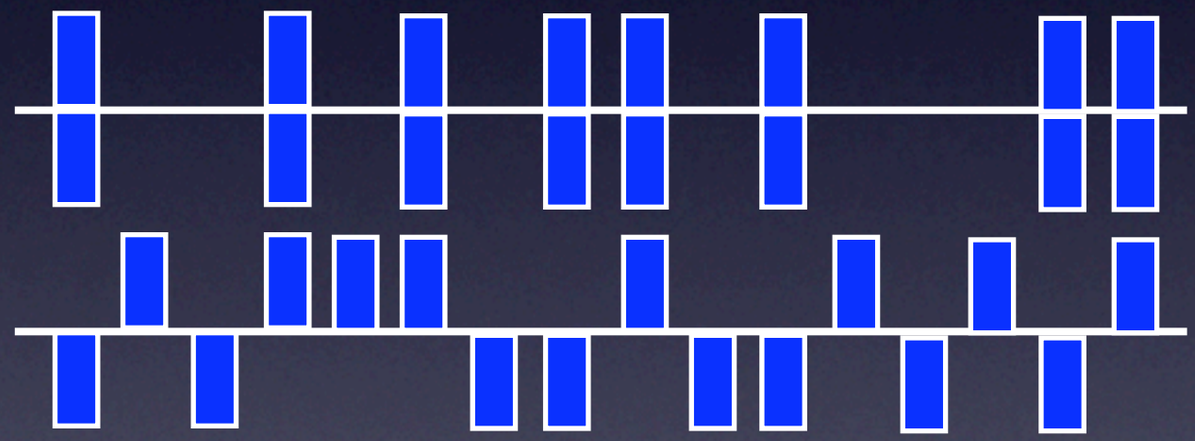
The Symmetry "Conjecture"

Vague version I: Quantum computers exponentially outperform classical computers because they can efficiently "exploit" symmetry

Vague version II: Permutation groups admit unitary representations which can be useful computationally only on quantum computers

Example

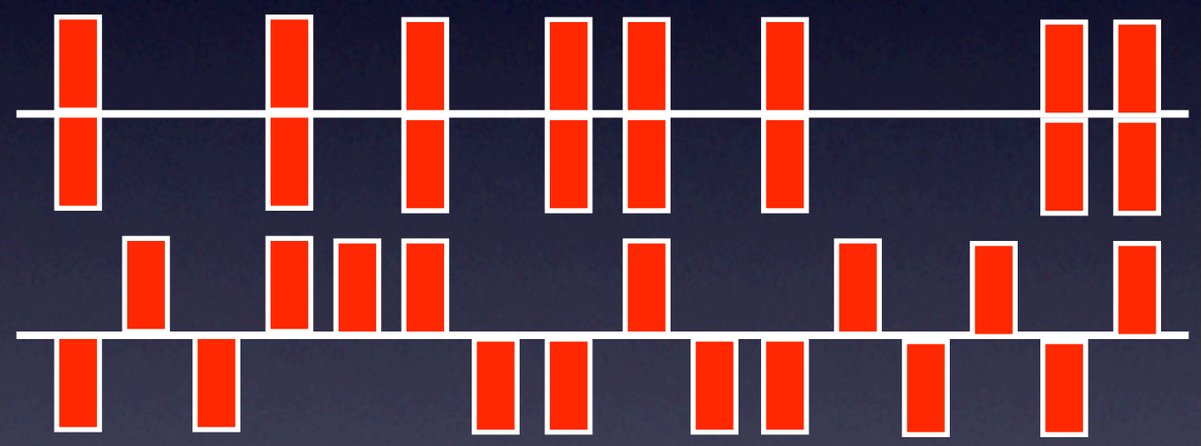
- Two probability distributions p and q over {1,2,..2N}
- Both are evenly distributed over subset size N
  - p equals q
  - p is complement of q



- Problem: distinguish two cases
- Classically: $\Omega(N^{1/2})$ copies of this distribution needed

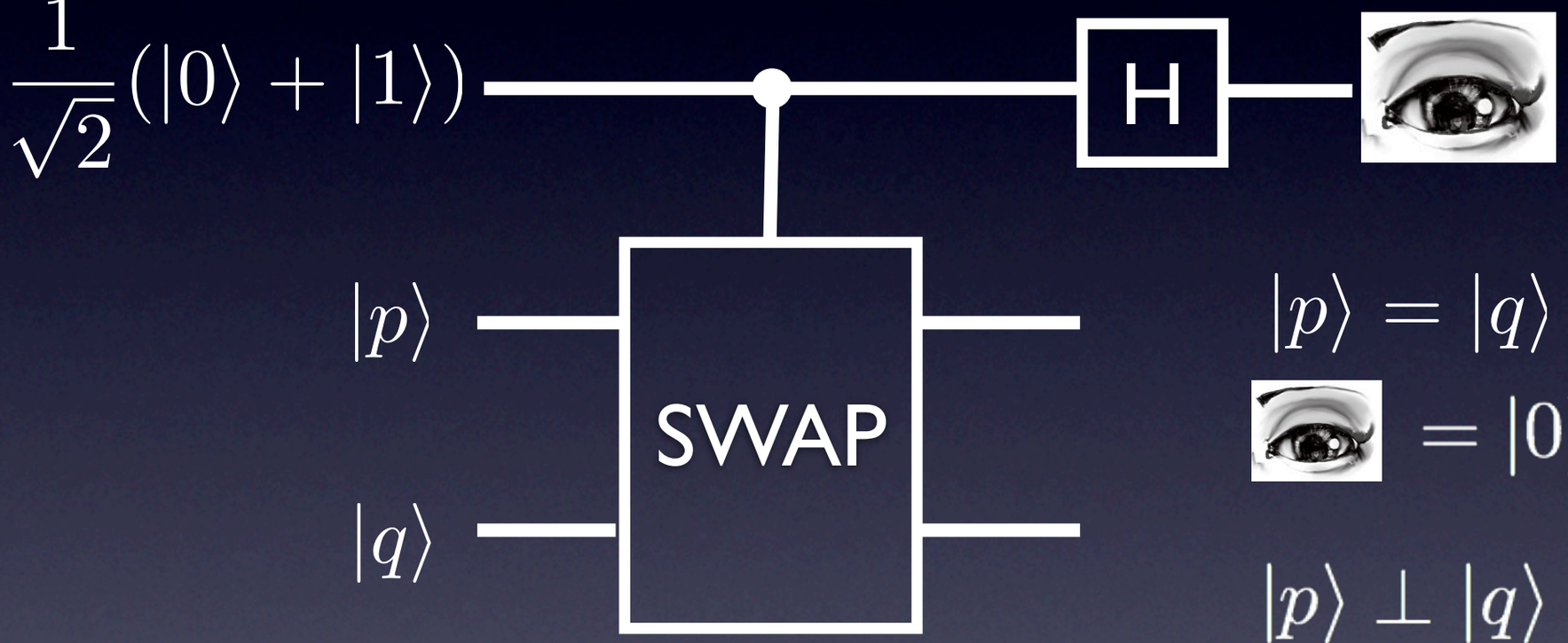
Example

- Two amplitude distributions p and q over {1,2,..2N}
- Both are evenly distributed over subset size N
  - p equals q
  - p is complement of q



- Problem: distinguish two cases
- Quantum: O(1) copies of distribution needed

The Swap Test



$|p\rangle = |q\rangle$

$= |0\rangle \quad 100\%$

$|p\rangle \perp |q\rangle$



$= |0\rangle \quad 50\%$

$= |1\rangle \quad 50\%$

What's this got to do with symmetry?

Symmetry



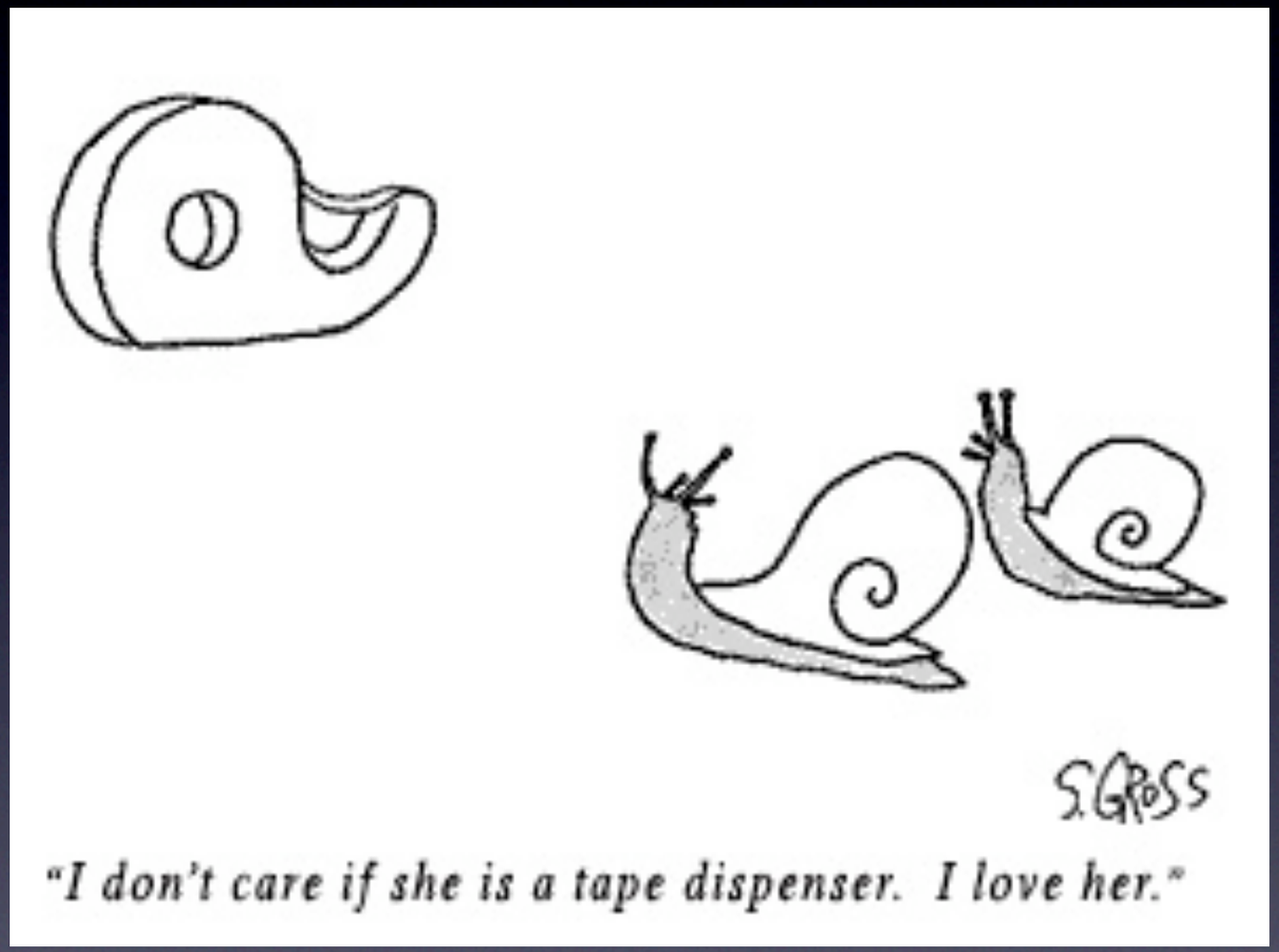
$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}^{\otimes n}$$

Under unitary basis change this becomes

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}^{\otimes n} \begin{matrix} \left.\begin{matrix} |00\rangle \\ \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \\ |11\rangle \end{matrix}\right\} \\ \left.\begin{matrix} \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \end{matrix}\right\} \end{matrix}$$

Classical probabilistic transforms are stochastic and cannot perform this basis change.

Cognitive Dissonance



Algorithms that offer exponential speedups
but don't seem to exploit symmetry

Simon's Problem

(is that nobody does what Simon says)

Given: A function with n bit strings as input and one bit as output

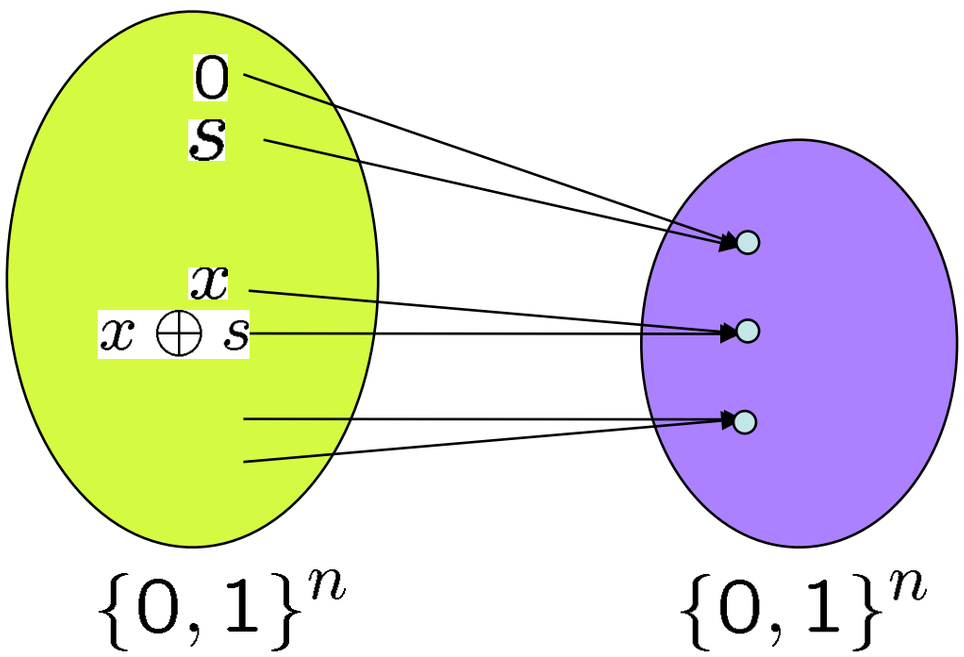
$$f : x \in \{0,1\}^n \rightarrow \{0,1\}^n$$

Promise: The function is guaranteed to satisfy

$$f(x) = f(y) \Leftrightarrow y = x \oplus s, s \neq 0$$

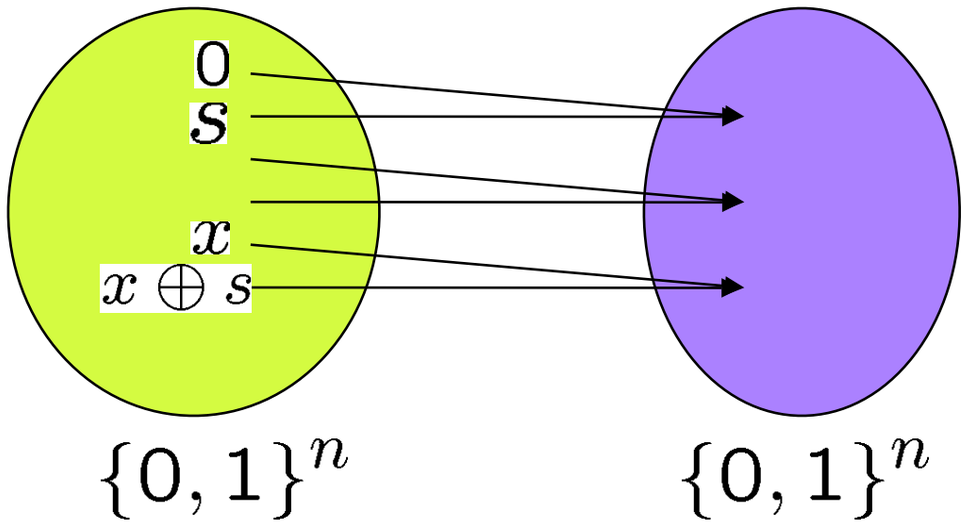
$$x \oplus s = (x_1 \oplus s_1, x_2 \oplus s_2, \ldots, x_n \oplus s_n)$$

Problem: Find the n bit string $s \neq 0$



Classical Simon's Problem

query pairs $(x_\alpha, f(x_\alpha))$



If we find $x_\alpha \neq x_\beta$ such that $f(x_\alpha) = f(x_\beta)$ then

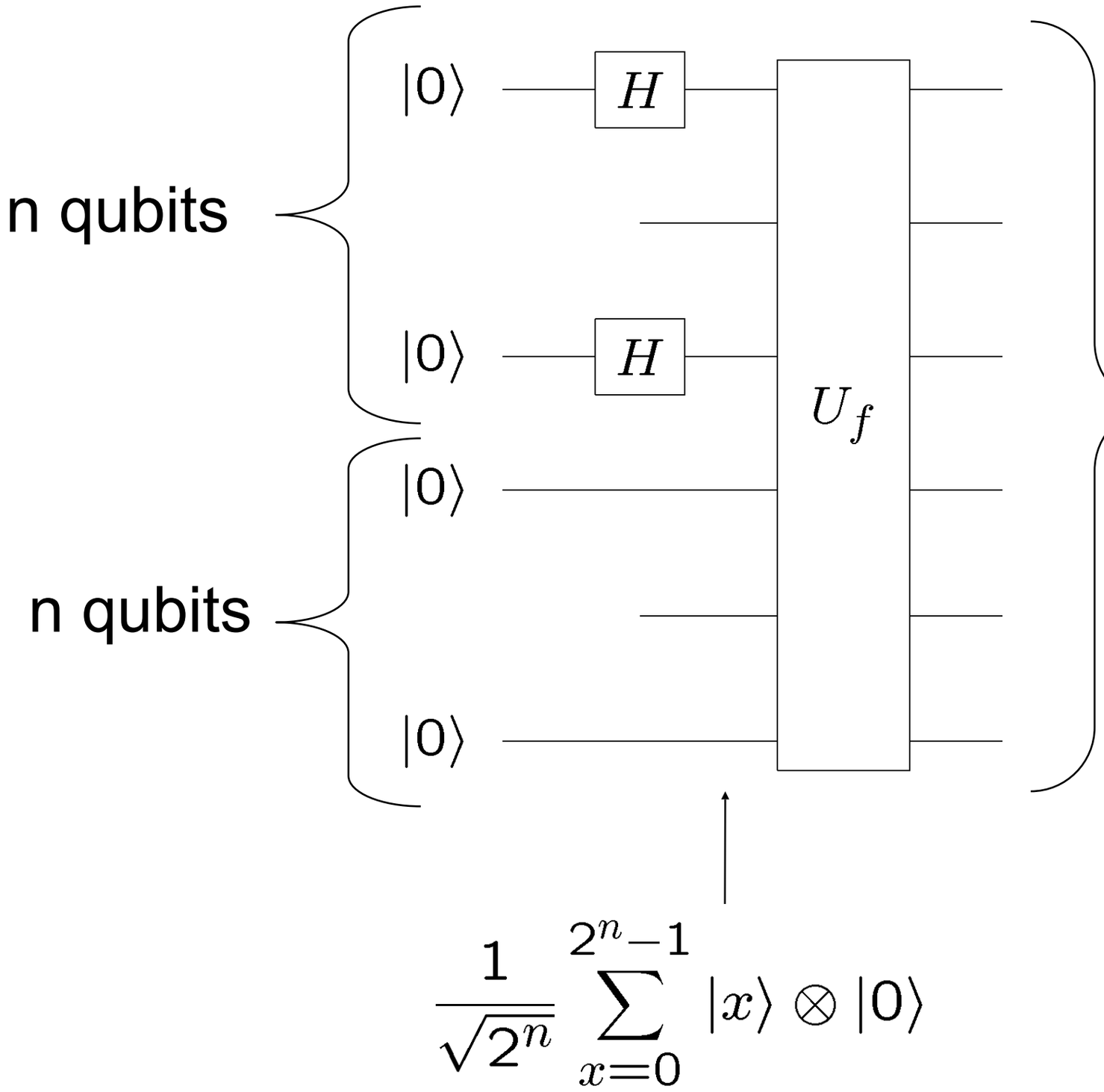
$$f(x_\alpha) = f(x_\beta) \Rightarrow x_\alpha = s \oplus x_\beta$$
$$s = x_\alpha \oplus x_\beta$$

Querying the function *m* times….

Probability of getting a matching pair $\approx \frac{\binom{m}{2}}{2^n} = \frac{m(m-1)}{2^{n+1}}$

Bounded error query complexity: $m = O\left(2^{\frac{n}{2}}\right)$

Quantum Simon's Problem



Quantum Simon's Problem

For random uniformly distributed $x \in \{0,1\}^n$

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle \otimes |f(x)\rangle \longrightarrow |\psi_s(x)\rangle = \frac{1}{\sqrt{2}}(|x\rangle + |x \oplus s\rangle)$$

$$\rho_s = \frac{1}{2^n} \sum_x |\psi_s(x)\rangle\langle\psi_s(x)|$$

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Symmetry:

$$R = X^{a_1} \otimes X^{a_2} \otimes \cdots X^{a_n} \qquad a_i \in \{0,1\}$$

$$R\rho_s R^\dagger = \rho_s \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Quantum Simon's Problem

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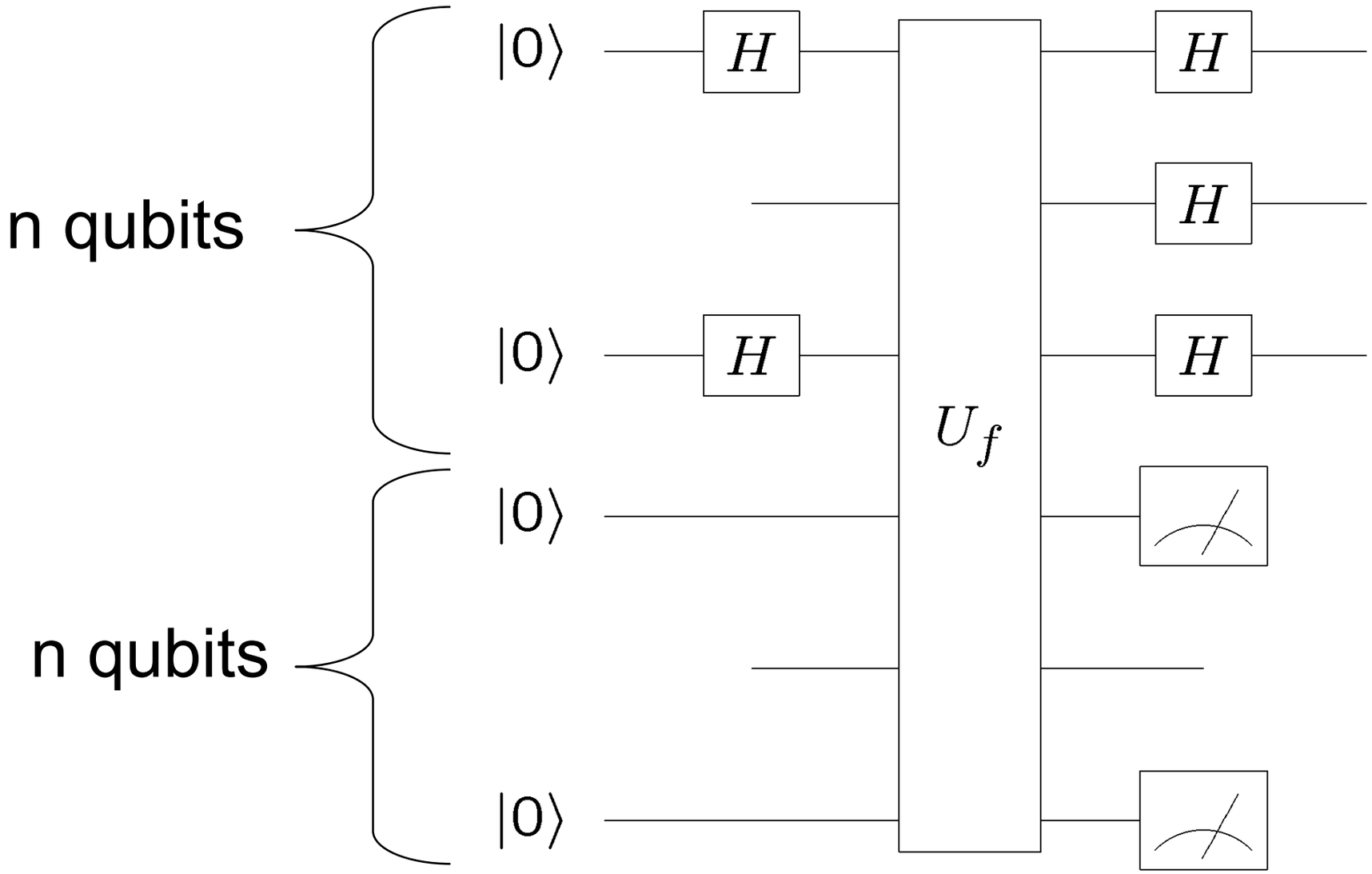
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Symmetry → Simultaneously diagonalize symmetry operators and symmetric operator

$$HZH^\dagger = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Quantum Simon's Problem



$$H^{\otimes n}\frac{1}{\sqrt{2}}(|x\rangle+|x\oplus s\rangle) = \frac{1}{\sqrt{2^{n+1}}}\sum_{y=0}^{2^n-1}((-1)^{y\cdot x}+(-1)^{y\cdot(x\oplus s)})|y\rangle$$

$$y\cdot x = y_1x_1 \oplus y_2x_2 \oplus \cdots \oplus y_nx_n$$

Quantum Simon's Problem

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$$= \frac{1}{\sqrt{2^{n+1}}}\sum_{y=0}^{2^n-1}(-1)^{y\cdot x}(1+(-1)^{y\cdot s})|y\rangle$$

Measuring this state, uniformly distributed random y such that

$$y \cdot s = 0 \mod 2$$

Quantum Simon's Problem

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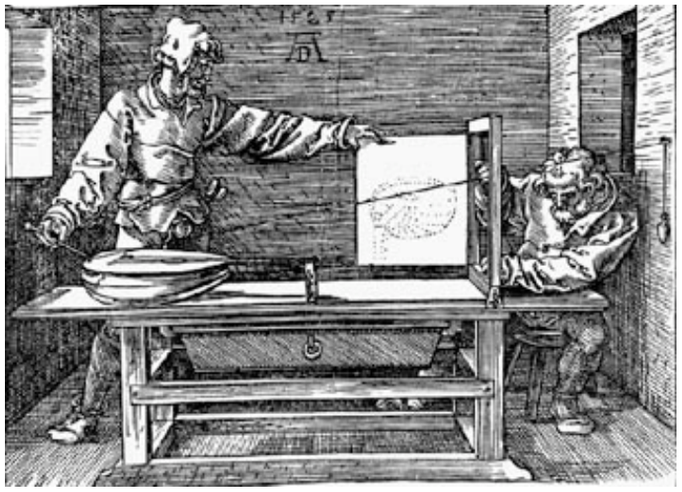
Multiple runs:

$$\begin{aligned} y_1\cdot s &= 0 \\ y_2\cdot s &= 0 \\ &\vdots \\ y_k\cdot s &= 0 \end{aligned}$$

random uniform y_i

If we obtain $n-1$ linearly independent equations of this form, we win (Gaussian elimination)

Moral



Symmetries

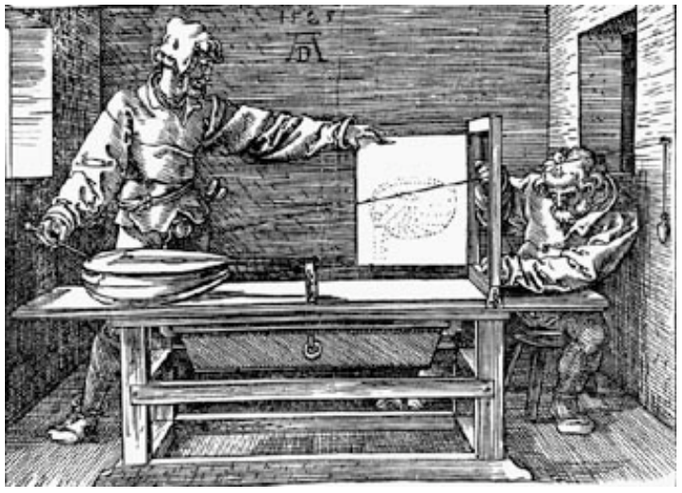
↓

Diagonalize with respect to symmetry

↓

Quantum algorithm

Moral



Symmetries

↓

Diagonalize with respect to symmetry

↓

Quantum algorithm

Quantum	Classical
$S\rho S^\dagger = \rho$	$S\vec{p} = \vec{p}$
ρ and S simultaneously diagonalized	What do we do?

But what about that Gaussian Elimination?

Representation Theory in 1 Slide

Group $\mathcal{G}$ (Set with binary operator which is closed, associative, has an identity, and inverses)

Unitary **representation** of group $T : \mathcal{G} \rightarrow U(d)$

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T_γ has no invariant subspace except full space and empty subspace

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$$T(g) = \bigoplus_i T_{\gamma_i}(g) = \bigoplus_\gamma I_{n_\gamma} \otimes T_\gamma(g)$$

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Schur's lemma:

$$T(g)\rho = \rho T(g)\ \forall g \implies \rho = \bigoplus_\gamma Q_\gamma \otimes I_{d_\gamma}$$

Simon's Problem Redux

Group $\mathcal{G} = \mathbb{Z}_2^n$ addition bitwise mod *2*

$$(x_1, \ldots, x_n) + (y_1, \ldots, y_n) = (x_1 + y_1, \ldots, x_n + y_n)$$

$$x_i, y_i \in \{0, 1\}$$

Representation of group

$$T((x_1, \ldots, x_n)) = X^{x_1} \otimes \cdots \otimes X^{x_2}$$

Irreps of group:

$$T_{(k_1,\ldots,k_n)}((x_1, \ldots, x_n)) = (-1)^{x_1 k_1 + \cdots + x_n k_n}$$

Unitary transform to direct product of irreps:

$$H^{\otimes n} T((x_1, \ldots, x_n)) H^{\otimes n\dagger} = \bigoplus_k T_{k_1,\ldots,k_n}((x_1, \ldots, x_n))$$

More Simon's Problem Redux

$$y_1 \cdot s = 0$$
$$y_2 \cdot s = 0$$
$$\vdots$$
$$y_k \cdot s = 0$$

random
uniform
y_i

are irrep labels

Gaussian elimination : adding y_i bitwise mod *2*

Symmetry interpretation?

Symmetry Interpretation

Clebsch-Gordan transform:

$$T_\alpha(g) \otimes T_\beta(g) = \bigoplus_\gamma I_{n_\gamma} \otimes T_\gamma(g)$$

$$\begin{aligned} & T_{(k_1,\ldots,k_n)}((x_1,\ldots,x_n)) \otimes T_{(l_1,\ldots,l_n)}((x_1,\ldots,x_n)) \\ &= (-1)^{x_1 k_1 + \cdots + x_n k_n} (-1)^{x_1 l_1 + \cdots + x_n l_n} \\ &= T_{(k_1+l_1,\ldots,k_n+l_n)}((x_1,\ldots,x_n)) \end{aligned}$$

So the “addition” of Gaussian elimination in Simon’s algorithm can be interpreted as a Clebsch-Gordan transform….

The Second Basis Change Learned In Quantum Mechanics:

Clebsch-Gordan Transform

Singlet:

$$|S=0,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle)$$

Triplet:

$$|S=1,m=1\rangle = |\uparrow\rangle \otimes |\uparrow\rangle$$

$$|S=1,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle + |\downarrow\rangle \otimes |\uparrow\rangle)$$

$$|S=1,m=-1\rangle = |\downarrow\rangle \otimes |\downarrow\rangle$$

What Use?

Collective Unitaries

$$|S=0,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle)$$

$$|S=1,m=1\rangle = |\uparrow\rangle \otimes |\uparrow\rangle$$

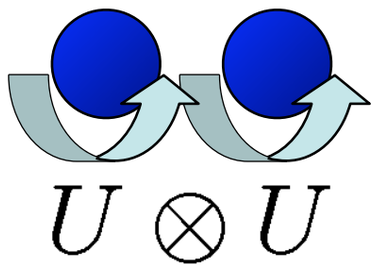
$$|S=1,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle + |\downarrow\rangle \otimes |\uparrow\rangle)$$

$$|S=1,m=-1\rangle = |\downarrow\rangle \otimes |\downarrow\rangle$$

the collective

a Division of Foundation 9 Entertainment

$$\left[\begin{array}{c|c} 1 & 0 \\ \hline 0 & U \end{array}\right]$$



$U \otimes U$

Collective Unitaries

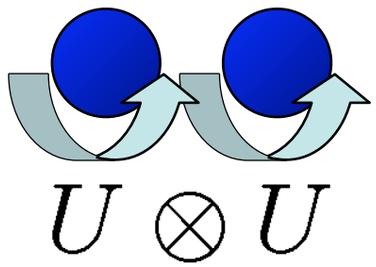
$$|S=0,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle)$$

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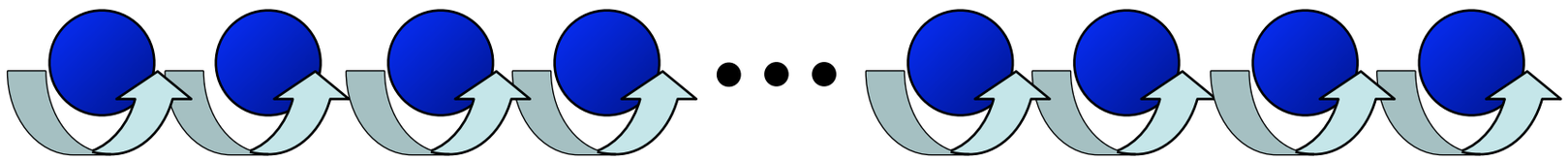
$$|S=1,m=-1\rangle = |\downarrow\rangle \otimes |\downarrow\rangle$$

$$\left[\begin{array}{c|c} 1 & 0 \\ \hline 0 & U \end{array}\right]$$

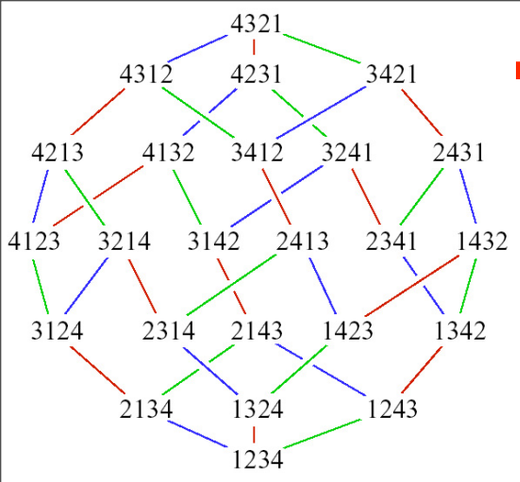


$U \otimes U$

Symmetry Under n Collective Unitary Rotations



$U^{\otimes n}$



The Symmetric Group

Transform Under SWAP

Singlet:

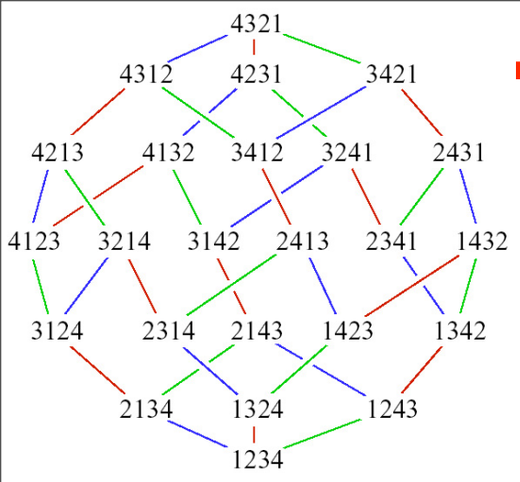
$$|S=0,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle) \quad \Big\} \quad -1$$

Triplet:

$$|S=1,m=1\rangle = |\uparrow\rangle \otimes |\uparrow\rangle$$

$$|S=1,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle + |\downarrow\rangle \otimes |\uparrow\rangle) \quad \Big\} \quad +1$$

$$|S=1,m=-1\rangle = |\downarrow\rangle \otimes |\downarrow\rangle$$



The Symmetric Group

Transform Under SWAP

Singlet:

$$|S=0,m=0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle) \quad \Big\} \quad -1$$

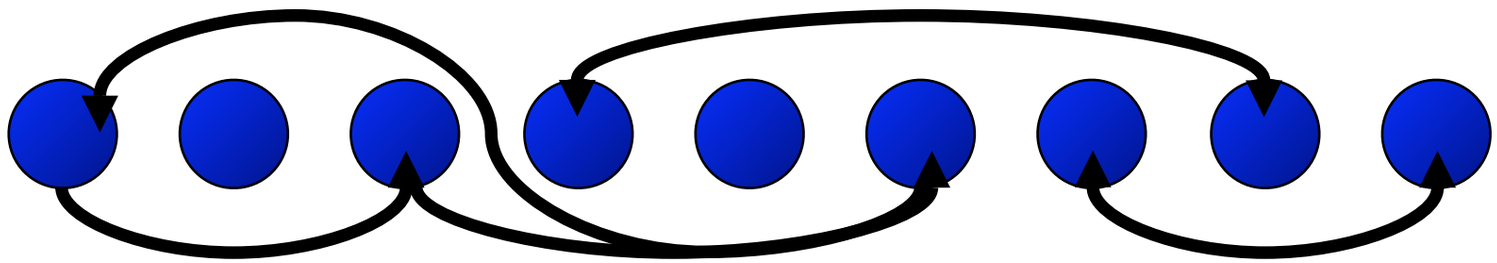
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$$|S=1,m=-1\rangle = |\downarrow\rangle \otimes |\downarrow\rangle$$

Symmetry Under Permutation of n Subsystems



Quantum Schur Transform

n d-dimensional systems (n qudits)



Young diagram

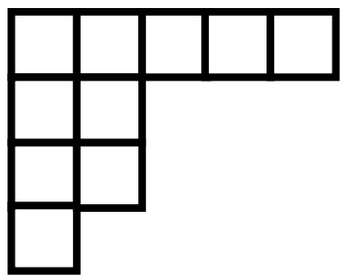
$\mathcal{S}_n$ irrep space

$SU(d)$ irrep space

Schur duality

$$[\mathbb{C}^d]^{\otimes n} \overbrace{\cong}^{\text{Schur}} \bigoplus_{\mathcal{Y}} S_{\mathcal{Y}} \otimes U_{\mathcal{Y}}$$

Young diagram



$$k_1 \geq k_2 \geq \cdots \geq k_d$$

$$\sum_{i=1}^{d} k_i = n$$

The Plethora of Uses of Schur

Optimal spectrum estimation

[M. Keyl and R. F. Werner. Phys. Rev. A, 2001]

Universal distortion-free entanglement concentration

[M. Hayashi and K. Matsumoto. Information Theory, 2004]

Universal compression with optimal overflow exponent

[M. Hayashi and K. Matsumoto. Phys. Rev. A, 2002]

Optimal quantum hypothesis testing

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Communicating without a shared reference frame

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Encoding into decoherence-free (noiseless) subsystems

[J. Kempe, D. Bacon, D. A. Lidar, and K. B. Whaley, Phys. Rev. A, 2001]

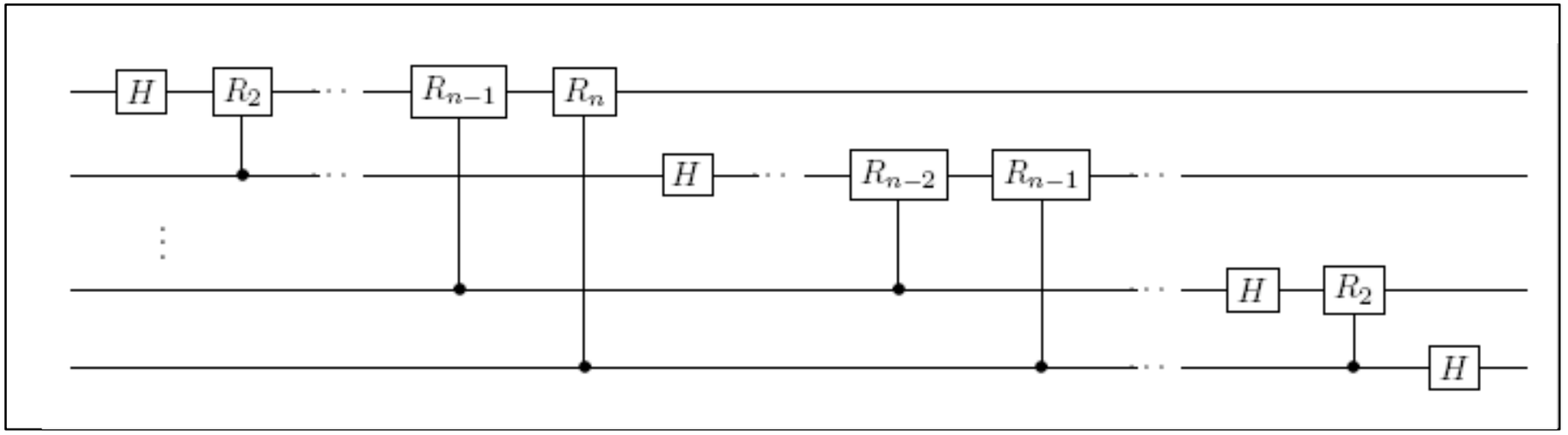
Simulating lattice gauge theories on a quantum computer

[T. Byrnes and Y. Yoshihisa, Phys. Rev. A 2006]

….

But Efficient?

QFT

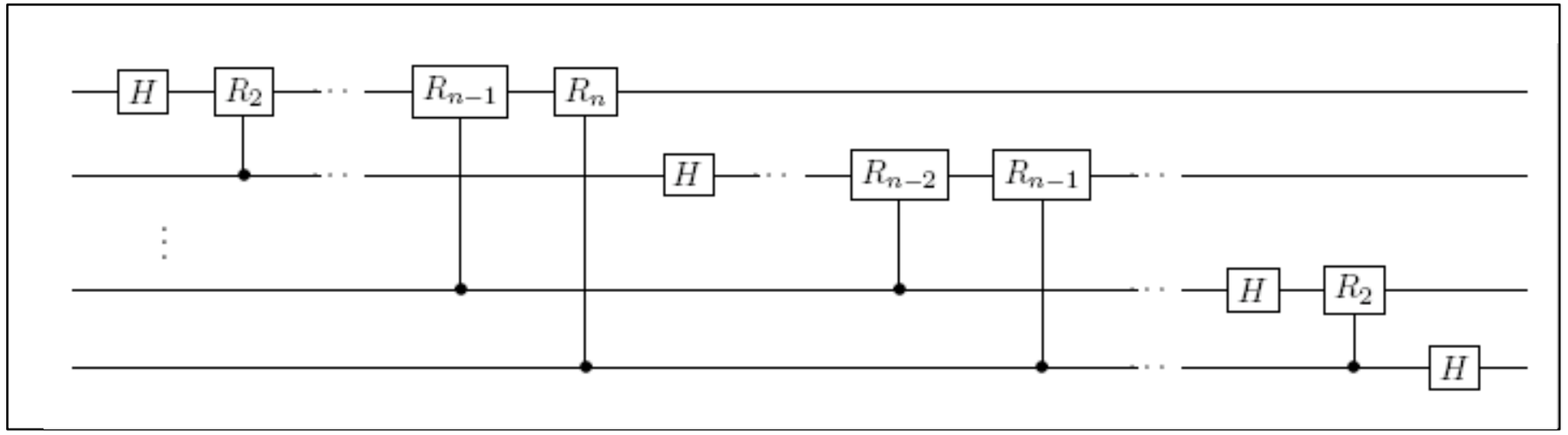


Schur

?

But Efficient?

QFT

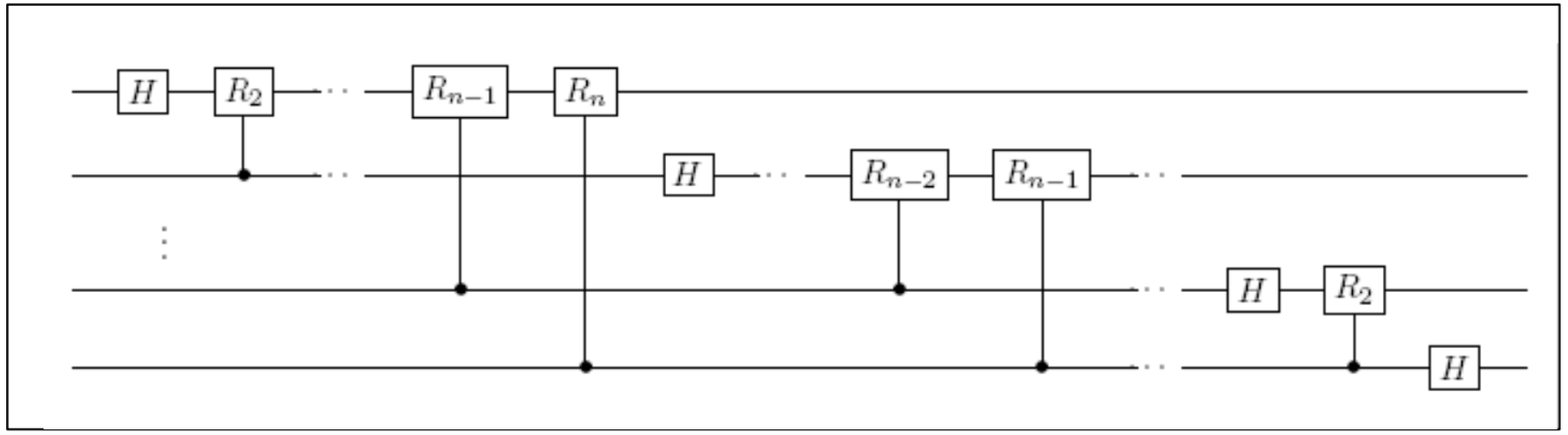


Schur



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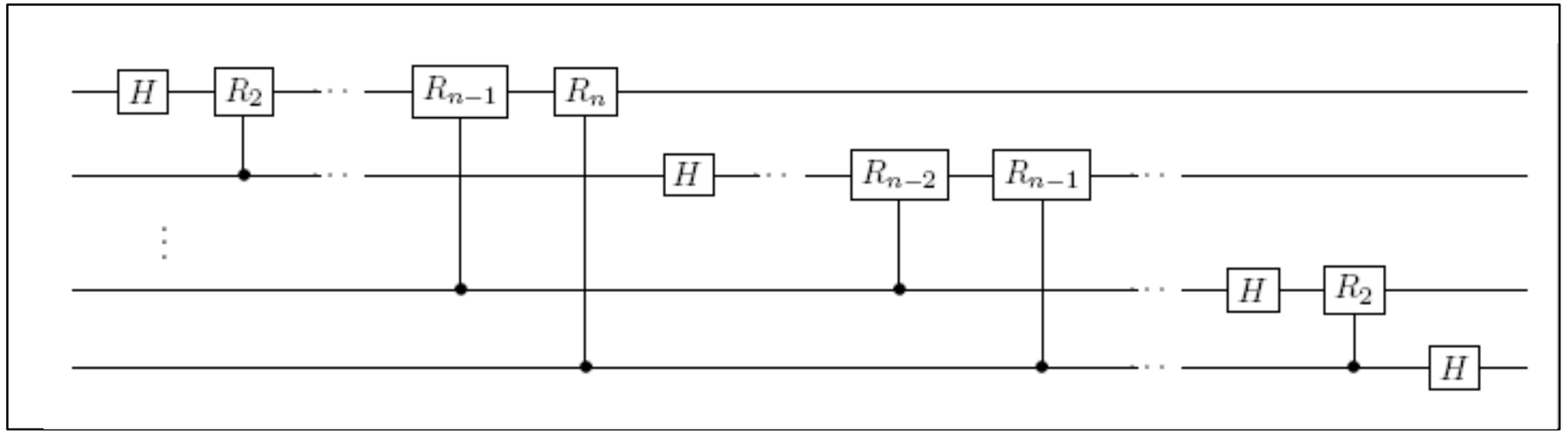
Schur



Who would need to efficiently calculate Clebsch-Gordan coefficients for SU(d)?

But Efficient?

QFT



Schur



Who would need to efficiently calculate Clebsch-Gordan coefficients for SU(d)?

[L. C. Biedenharn and J. D. Louck, Commun. Math. Phys. 1968]

Efficient Schur

[D. Bacon, I.L. Chuang, A. Harrow, Phys. Rev. Lett. 2006]
[D. Bacon, I.L. Chuang, A. Harrow, SODA 2007]

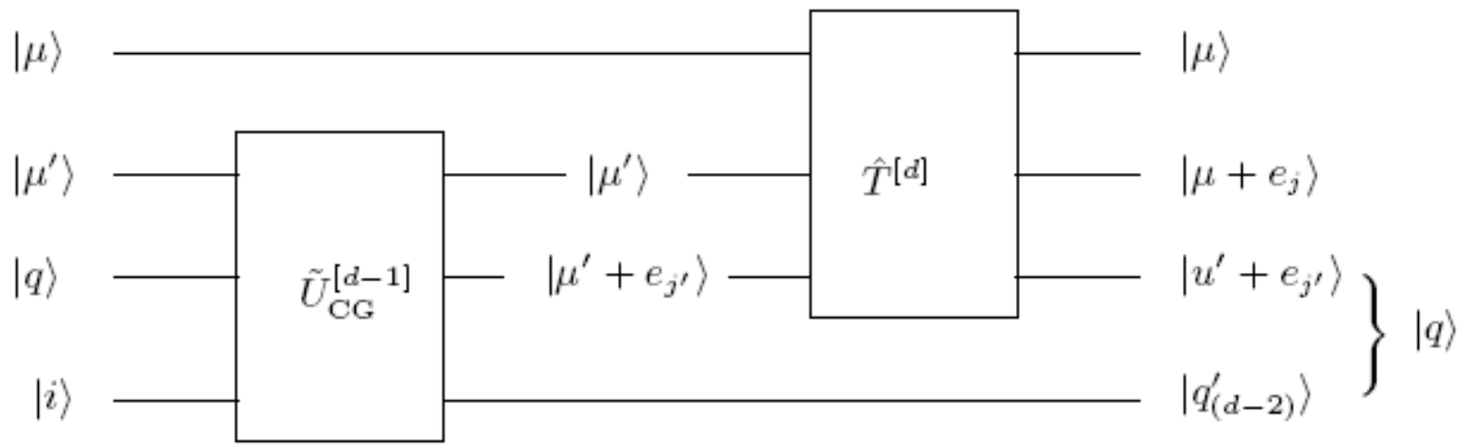


FIG. 4: The $\mathcal{U}_d$ CG transform, $\mathbf{U}^{[d]}_{\mathrm{CG}}$, is decomposed into a $\mathcal{U}_{d-1}$ CG transform $\tilde{U}^{[d-1]}_{\mathrm{CG}}$ (see Eq. (54)) and a reduced Wigner operator $\hat{T}^{[d]}$. In Fig. 5 we show how to reduce the reduced Wigner operator to a $d \times d$ matrix conditioned on μ and $\mu' + e_{j'}$.

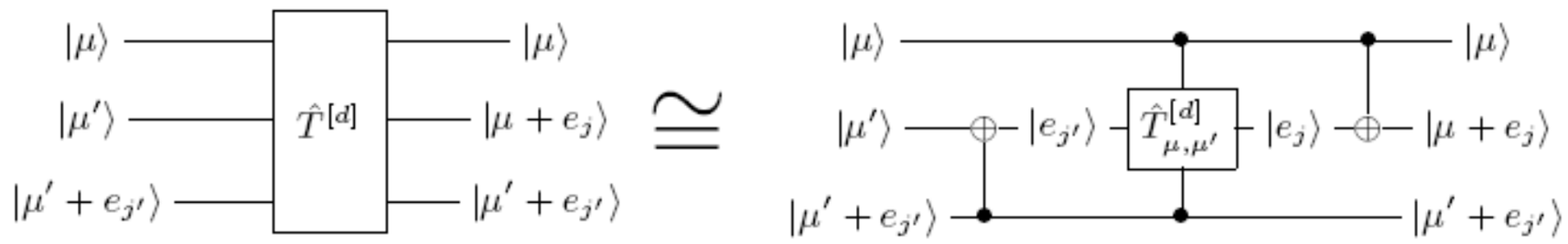


FIG. 5: The reduced Wigner transform $\hat{T}^{[d]}$ can be expressed as a $d \times d$ rotation whose coefficients are controlled by μ and $\mu' + e_{j'}$.

Schur transform to accuracy ϵ : $poly\left(log\frac{1}{\epsilon}, n, d\right)$ time

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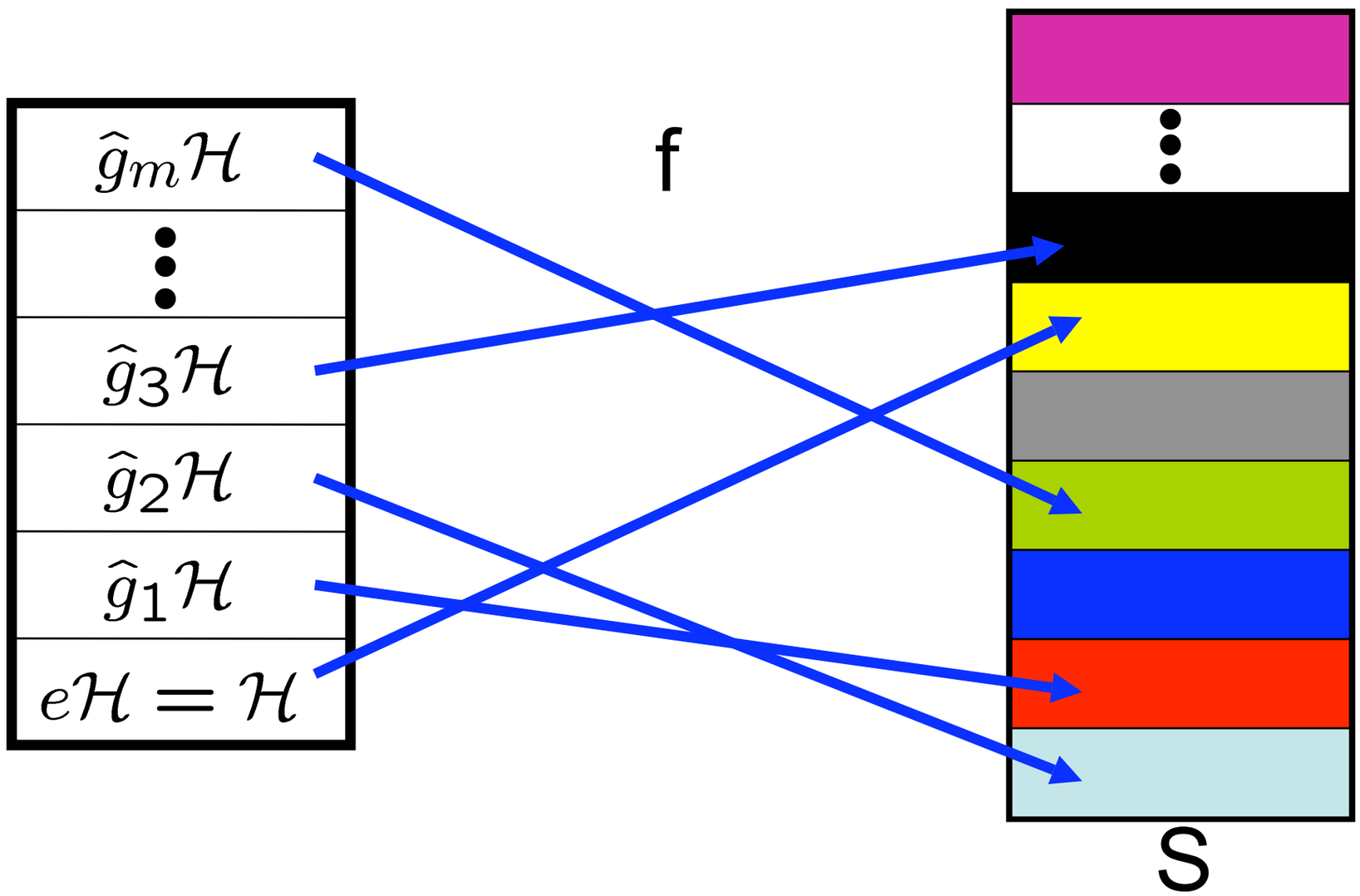
….

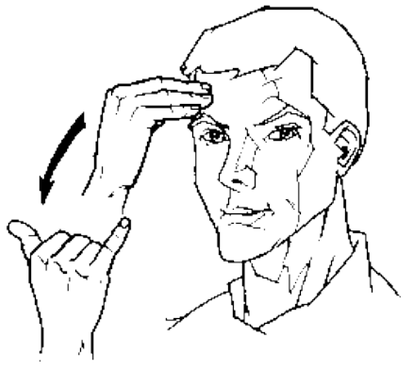
Tractable

But Quantum Algorithms?

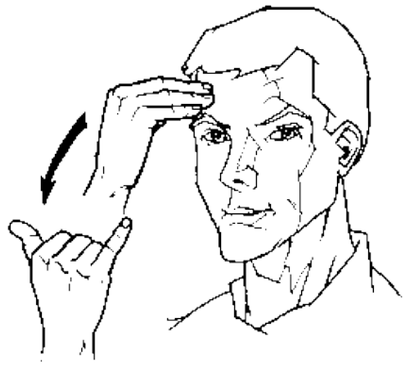
The Hidden Subgroup Problem

Setup: Fix a group $\mathcal{G}$ (known) and a subgroup $\mathcal{H} \subset \mathcal{G}$ (unknown). Given query access to a function f : $\mathcal{G} \rightarrow$ S that is constant and distinct on left (right) cosets of $\mathcal{H}$ in $\mathcal{G}$. We say that f hides $\mathcal{H}$.
Problem: Find a generating set for $\mathcal{H}$ ("identify $\mathcal{H}$").
Efficient: If runs in time polylog(| $\mathcal{G}$ |)



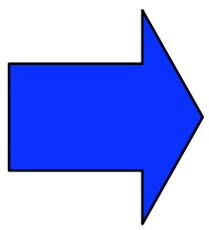


Why the Fuss?



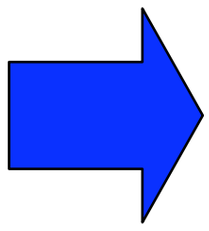
Solved Problems (Shor 1994)

efficient algorithm for HSP over $\mathbb{Z}_N$



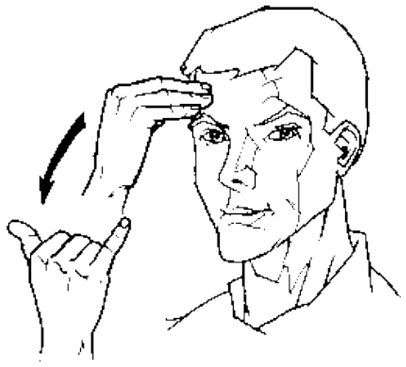
efficient algorithm for factoring whole numbers

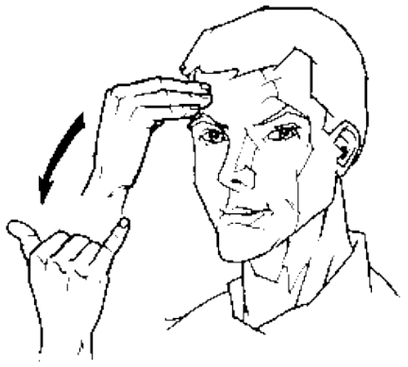
efficient algorithm for HSP over $\mathbb{Z}_N \times \mathbb{Z}_N$



efficient algorithm for the discrete logarithm problem

Why the Fuss?





Solved Problems (Shor 1994)

efficient algorithm for HSP over $\mathbb{Z}_N$ → efficient algorithm for factoring whole numbers

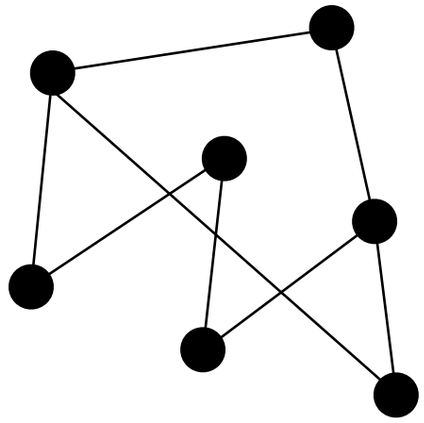
efficient algorithm for HSP over $\mathbb{Z}_N \times \mathbb{Z}_N$ → efficient algorithm for the discrete logarithm problem

Open Problems

efficient algorithm for HSP over $\mathcal{S}_n \wr \mathbb{Z}_2$ (or $\mathcal{S}_{2n}$) → efficient algorithm for the graph isomorphism problem

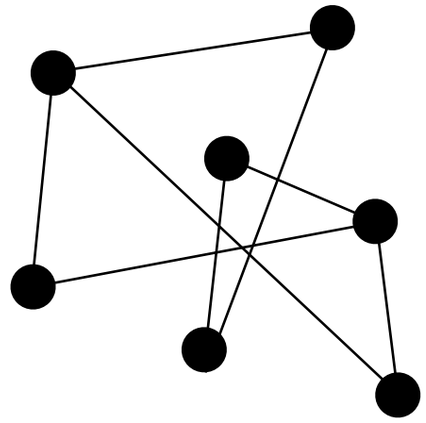
efficient "standard" quantum algorithm for HSP over dihedral group $\mathcal{D}_N$ → efficient algorithm for the poly(*n*) unique shortest vector in a lattice problem

Graph Isomorphism



adjacency matrix A_1

$G_1 = (V_1, E_1)$



adjacency matrix A_2

$G_2 = (V_2, E_2)$

Problem: Is there a 1-to-1 mapping p from the vertices V_1 to the vertices V_2 such that the edges are respected, $p(E_1) = E_2$?

Mapping to the HSP:

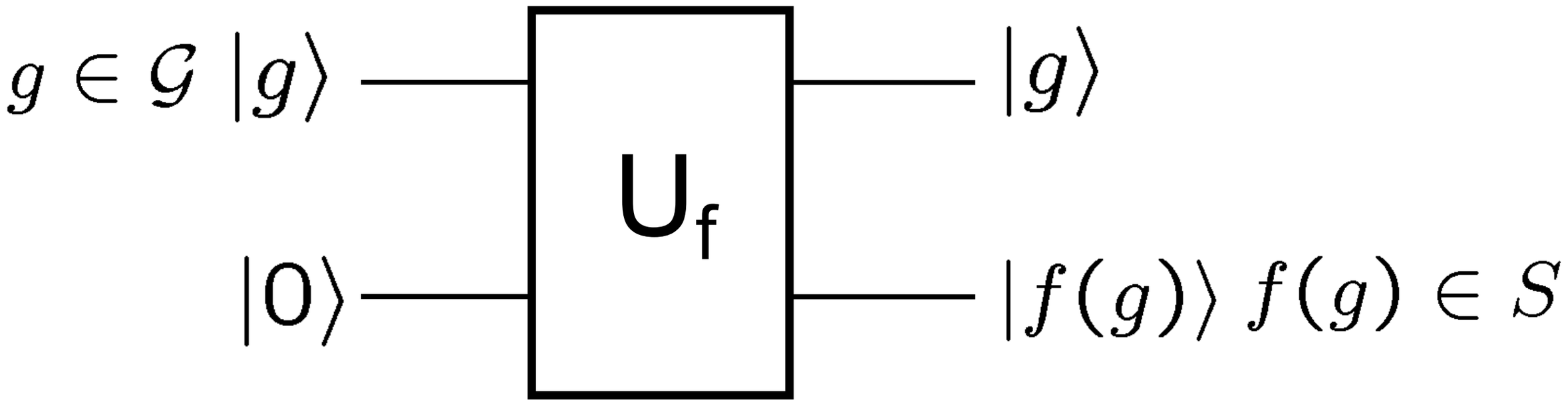
$$f : \mathcal{S}_{2n} \rightarrow \mathbb{Z}_{n(2n-1)}$$

$$f(\pi) = \pi(A_1 \oplus A_2)$$

If graphs are not isomorphic, hidden subgroup will not mix the graphs. If they are isomorphic, they will mix the graphs.

Standard Query Model

Oracle:



Query in equal superposition over group elements

$$\frac{1}{\sqrt{|\mathcal{G}|}} \sum_{g \in \mathcal{G}} |g\rangle|0\rangle \rightarrow \sum_{g \in \mathcal{G}} \frac{1}{\sqrt{|\mathcal{G}|}} |g\rangle|f(g)\rangle$$

$$\rho_{\mathcal{H}} = \frac{|\mathcal{H}|}{|\mathcal{G}|} \sum_{g \in \hat{\mathcal{G}}} |g\mathcal{H}\rangle\langle g\mathcal{H}| \quad \text{hidden subgroup state}$$

You Down With Symmetry?

$$\rho_{\mathcal{H}} = \frac{|\mathcal{H}|}{|\mathcal{G}|} \sum_{g \in \hat{\mathcal{G}}} |g\mathcal{H}\rangle\langle g\mathcal{H}| \quad \text{hidden subgroup state}$$

Symmetry under left regular representation:

$$D_L(g)|g'\rangle = |gg'\rangle \qquad D_L(g)\rho_{\mathcal{H}} D_L(g^{-1}) = \rho_{\mathcal{H}}$$

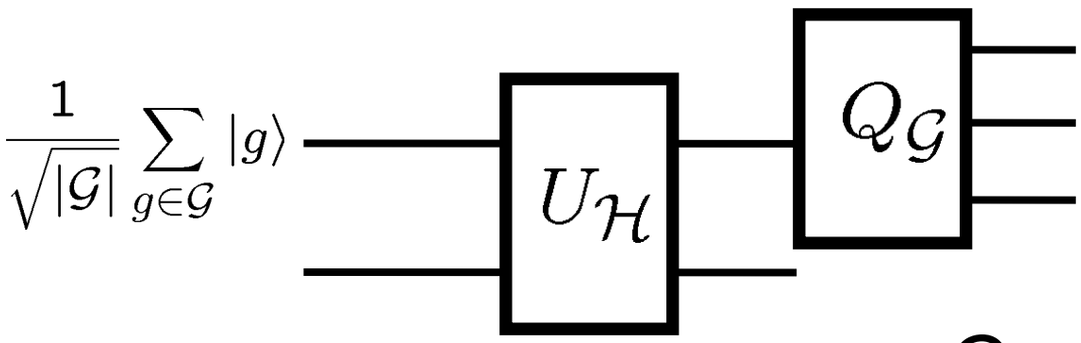
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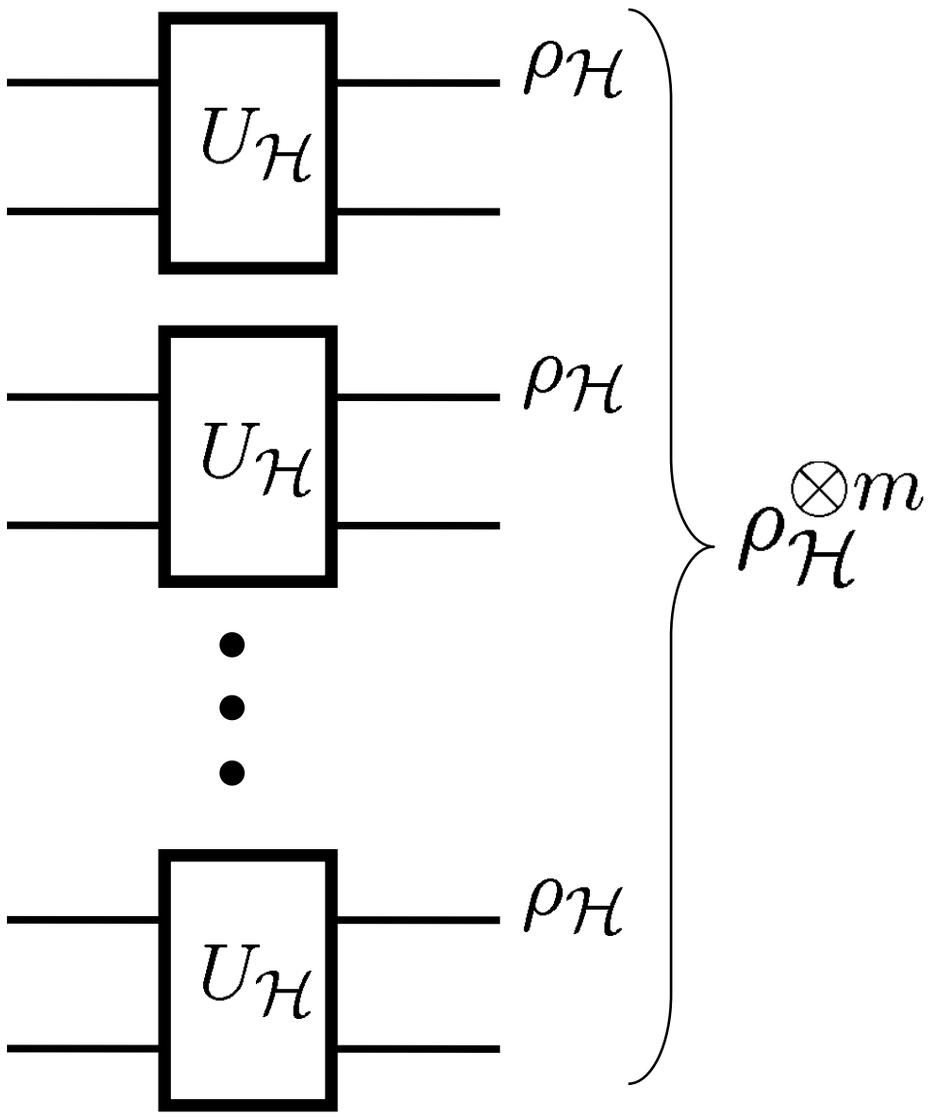
Fourier transform over finite group $\mathcal{G}$



Can we solve HSP this way?

Multicopy



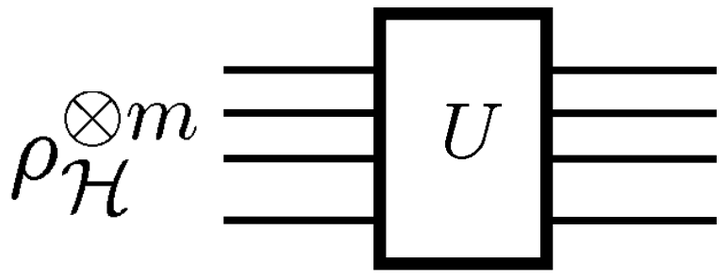


Can we make a measurement on $m < O(\log |\mathcal{G}|)$ copies and learn the subgroup?

In general (and for important case of symmetric group) **NO**

[Moore, Russell, Hallgren, Roetteler, Sen 05]

Required:



You Down With Symmetry?

$$\rho_{\mathcal{H}} = \frac{|\mathcal{H}|}{|\mathcal{G}|} \sum_{g \in \widehat{\mathcal{G}}} |g\mathcal{H}\rangle\langle g\mathcal{H}| \quad \text{hidden subgroup state}$$

Symmetry under left regular representation:

$$D_L(g)|g'\rangle = |gg'\rangle \qquad D_L(g)\rho_{\mathcal{H}} D_L(g^{-1}) = \rho_{\mathcal{H}}$$

Symmetry of two copies?

$$\rho_{\mathcal{H}} \otimes \rho_{\mathcal{H}}$$

Permutation symmetry….measure the Young Diagram?

Nope [Childs, Harrow, Wocjan 06]

Heisenberg Group

Semidirect product group $\mathbb{Z}_p^2 \rtimes \mathbb{Z}_p$



Elements of the group:

$$\left(\begin{pmatrix} x \\ y \end{pmatrix}, b \right) \quad x, y, b \in \mathbb{Z}_p$$

Multiplication rule:

$$\left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, b_1 \right) \left(\begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, b_2 \right) =$$

$$\left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^{b_1} \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, b_1 + b_2 \right)$$

From the play “Copenhagen” Margrethe Bohr to Heisenberg:

> “Your talent is for ski-ing too fast for anyone to see where you are. For always being in more than one position at a time, like one of your particles.”

Multicopy Success!

Efficient algorithm for the Heisenberg HSP: $\mathbb{Z}_p^2 \rtimes \mathbb{Z}_p$

[D. Bacon, A.M. Childs, and W. van Dam FOCS 2005]

PGM/optimal measurement approach:

$$\rho_{\mathcal{H}} \qquad Pr(succ) = O\left(\frac{1}{p}\right)$$

$$\rho_{\mathcal{H}} \otimes \rho_{\mathcal{H}} \qquad Pr(succ) = O(1)$$

Efficient algorithm for the Heisenberg HSP:

1. PGM is optimal and succeed w.h.p.
2. How to efficiently implement the PGM?

But Why Does It Work?

Clebsch-Gordan Transform To The Rescue?

Problem:

$\rho_{\mathcal{H}} \otimes \rho_{\mathcal{H}}$ does not have a Clebsch-Gordan related symmetry



Conjugate Subgroups

Two subgroups, $\mathcal{H}_1$ and $\mathcal{H}_2$, are conjugate if there exists an element of the group $g \in \mathcal{G}$ such that

$$\mathcal{H}_1 = \{ghg^{-1}, h \in \mathcal{H}_2\}$$

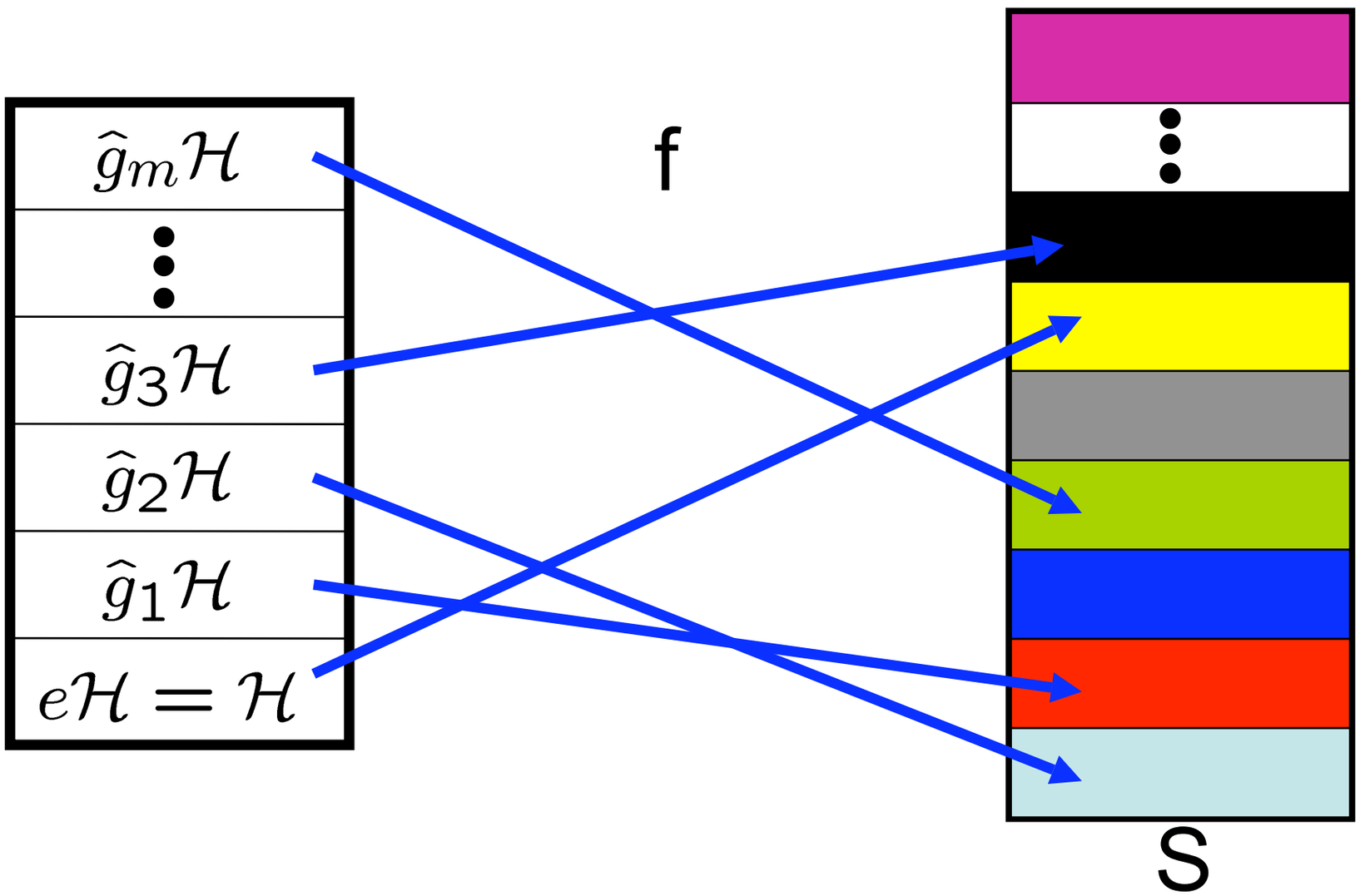
This is an equivalence relation among the subgroups of $\mathcal{G}$

$$\mathcal{H}_i \equiv_{\mathcal{G}} \mathcal{H}_j$$

ir: voy, vas, va, vamos, van

Hidden Subgroup Conjugacy Problem

Setup: Fix a group $\mathcal{G}$ (known) and a subgroup $\mathcal{H} \subset \mathcal{G}$ (unknown). Given query access to a function f : $\mathcal{G} \rightarrow$ S that is constant and distinct on left (right) cosets of $\mathcal{G}$ in $\mathcal{H}$. We say that f hides $\mathcal{H}$.
Problem: identify the conjugate subgroup class of $\mathcal{H}$.
Efficient: If runs in time polylog(| $\mathcal{G}$ |)



Symmetries of HSCP States

Hidden subgroup conjugate state

$$\rho_{[\mathcal{H}]} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \rho_{g\mathcal{H}g^{-1}}$$

Hidden subgroup conjugate states

$$\rho_{[\mathcal{H}]}^{\otimes m} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} (\rho_{g\mathcal{H}g^{-1}})^{\otimes m}$$

In context of optimal measurement method (PGM)

[e.g. Ip 03, Bacon, Childs, and van Dam 05]

this translates into extra symmetry of the optimal measurement

$$\rho_{[\mathcal{H}]}^{\otimes m} = \bigoplus_{\mu_1,\ldots,\mu_m} \left(I_D \otimes \bigoplus_{\mu} \left(R_{[\mathcal{H}]} \otimes I_{d_\mu} \right) \right)$$

Symmetries of HSCP States

What is the extra symmetry of the multiple HSCP states?

Clebsch-Gordan transform for the regular representation of finite groups:

$$D_R(g)^{\otimes m} = \left[\bigoplus_\mu (I_{d_\mu} \otimes D_\mu(g))\right]^{\otimes m} \qquad D_R(g)|g'\rangle = |g'g^{-1}\rangle$$

$$D_{\mu_1}(g) \otimes \cdots \otimes D_{\mu_m}(g) = \bigoplus_\mu \left(I_{n_\mu(\mu_1,\ldots,\mu_m)} \otimes D_\mu(g)\right)$$

Relationships

How do the hidden subgroup problem (HSP) and the hidden subgroup conjugacy problem (HSCP) differ?

1. If group is abelian, HSCP=HSP

All subgroups are only conjugate to themselves

$$g\mathcal{H}g^{-1} = \mathcal{H}$$

2. If subgroups are normal, HSCP=HSP

All subgroups are only conjugate to themselves

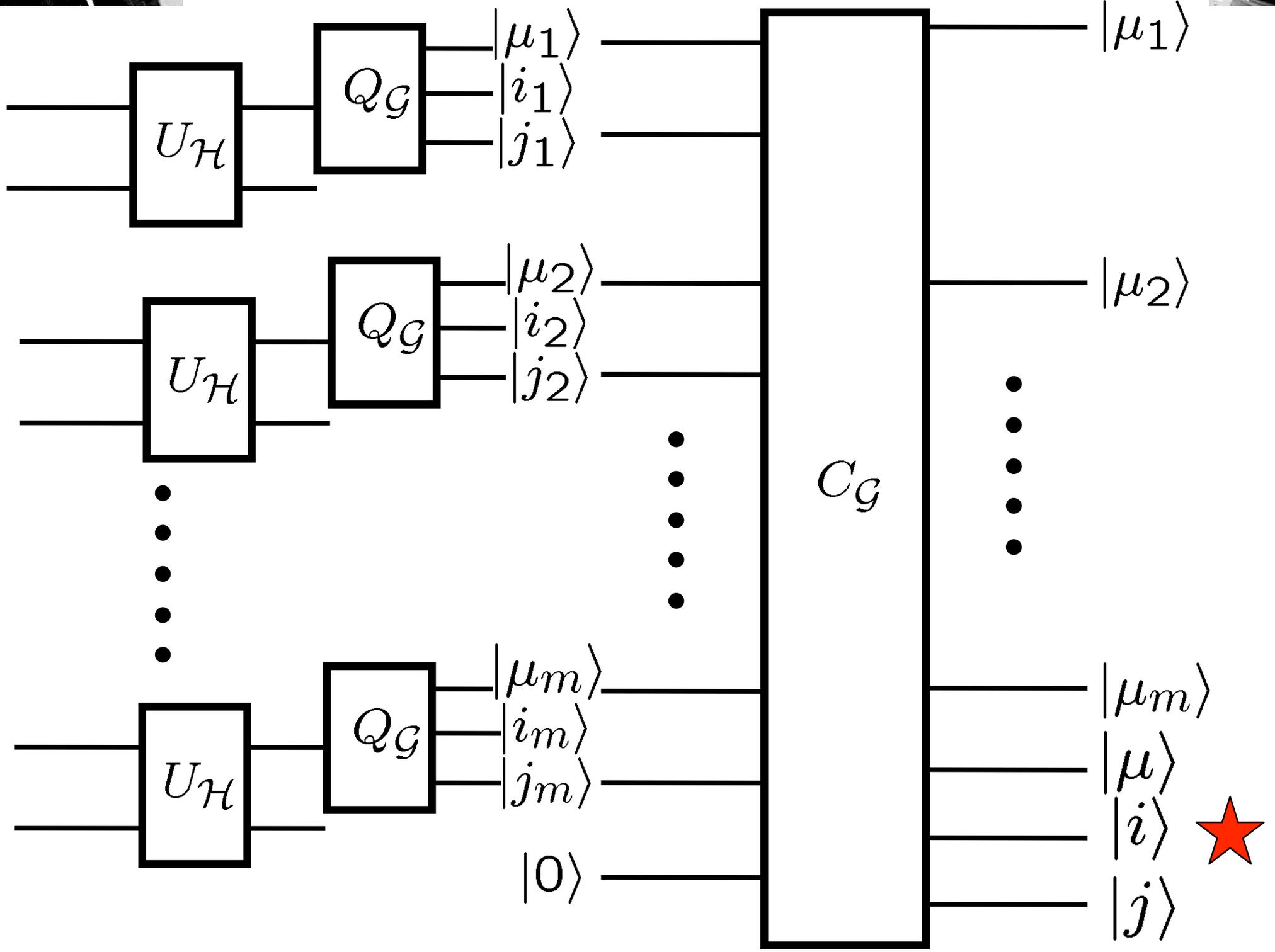
$$g\mathcal{H}g^{-1} = \mathcal{H}$$

These are two cases where we know quantum computers can efficiently solve the HSP. Coincidence?



General Circuit

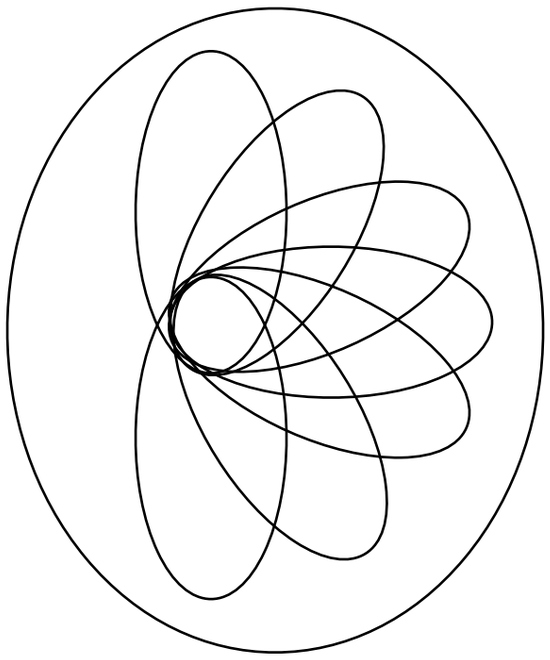




Heisenberg HSCP and HSP

[Bacon, arXiv:quant-ph/0612107]

Heisenberg HSCP $\Rightarrow$ Heisenberg HSP



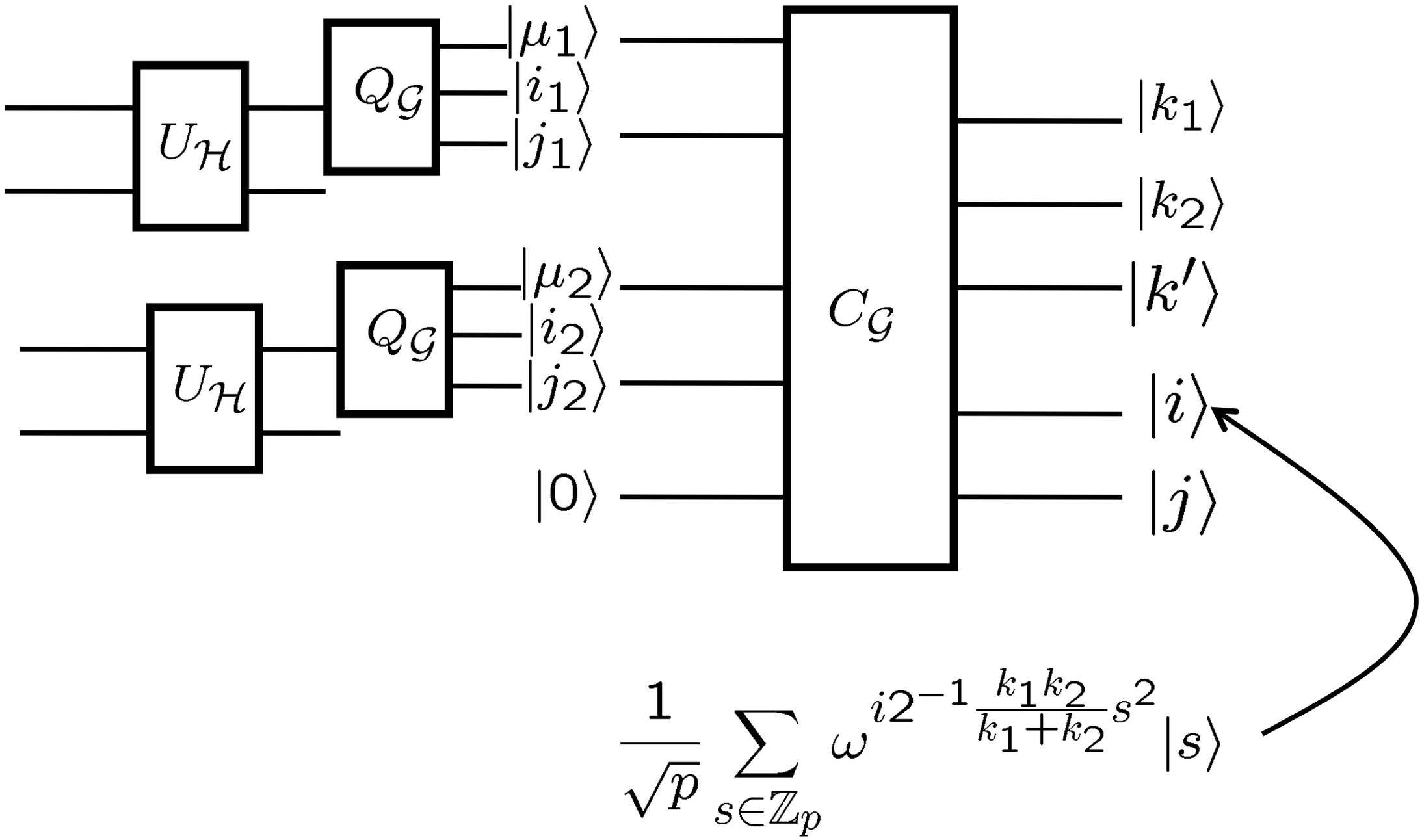
Conjugate subgroups are all normal subgroups of a subgroup of Heisenberg group

Identifying which conjugate subgroup identifies this subgroup

Thus solving HSCP implies solving HSP (use normal subgroup algorithm)

Heisenberg HSCP, CG Way

[Bacon, arXiv:quant-ph/0612107]



i labels set of conjugate subgroups

Heisenberg HSCP, CG Way

[Bacon, arXiv:quant-ph/0612107]

$$\frac{1}{\sqrt{p}} \sum_{s \in \mathbb{Z}_p} \omega^{i 2^{-1} \frac{k_1 k_2}{k_1 + k_2} s^2} |s\rangle$$

i can be found by performing (which can be done with high prob)

$$|t^2\rangle \rightarrow \frac{1}{\sqrt{2}}(|t\rangle + |-t\rangle)$$

and performing a QFT

Clebsch-Gordan transform over Heisenberg group is related to “symmetry” which allows for efficient quantum algorithm for the Heisenberg HSP.

Heisenberg HSP Lesson

Clebsch-Gordan transform can be used to help solve HSPs

Other groups where this works?

Krovi and Roetteler: Weyl-Heisenberg Groups

Bacon: Hidden symmetries of multiple squares $\mathcal{D}_4^n$

Krovi and Roetteler: some automorphisms can be used to change irrep labels...this allows for a two step sieve algorithm

Moore, Russell: sieving on irrep labels cannot be the strategy taken for the symmetric group

Fin

Clebsch-Gordan transform reveals essential symmetry of efficiently solvable nonabelian hidden subgroup problems….Can they lead us further, the $\$10^9$ question

funded by

ARO NSA NSF DARPA

Quantum Computing Peoples

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Undergraduates

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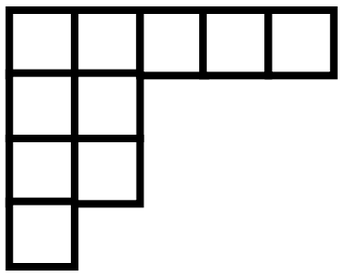
William Johnson (CSE)

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Uses of Schur

Young diagram



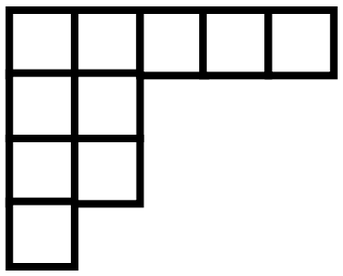
$$k_1 \geq k_2 \geq \cdots \geq k_d$$

$$\sum_{i=1}^{d} k_i = n$$

What does this remind you of?

Uses of Schur

Young diagram



$$k_1 \geq k_2 \geq \cdots \geq k_d$$

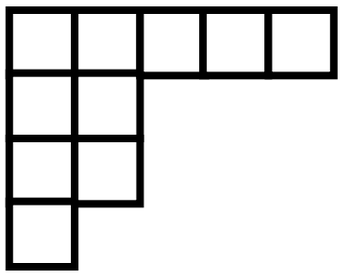
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What does this remind you of?

$\frac{k_i}{n}$ like probabilities like the spectrum of a density operator

Uses of Schur

Young diagram



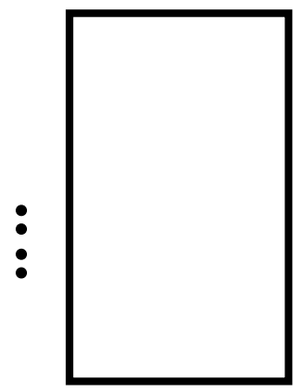
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$\rho^{\otimes n}$



Young diagram ← Optimal spectrum estimate

[Keyl and Werner. PRA, 2001]

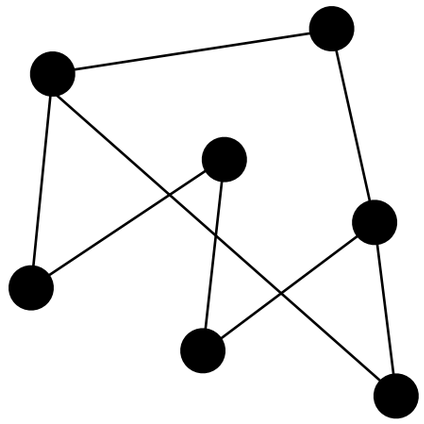
$\mathcal{S}_n$ irrep space

$SU(d)$ irrep space

You Down With Symmetry?

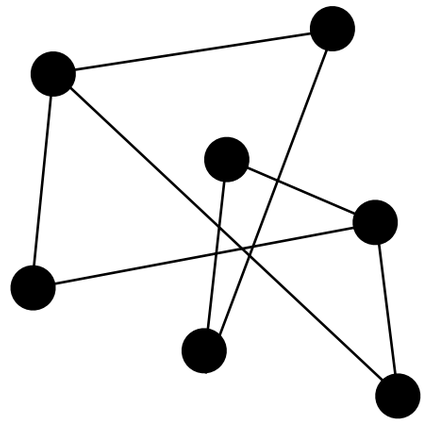
- And now, I will, for symmetries sake give the entire talk in reverse.

Graph Isomorphism



adjacency matrix A_1

$G_1 = (V_1, E_1)$



adjacency matrix A_2

$G_2 = (V_2, E_2)$

Problem: Is there a 1-to-1 mapping p from the vertices V_1 to the vertices V_2 such that the edges are respected, $p(E_1) = E_2$?

Mapping to the HSP:

$$f : \mathcal{S}_n \wr \mathcal{S}_2 \rightarrow \mathbb{Z}_{n(2n-1)}$$

$$f((\pi_1, \pi_2, \tau)) = \tau(\pi_1(A_1) \oplus \pi_2(A_2))$$

Notice now we are only worried if hidden subgroup has a nontrivial τ